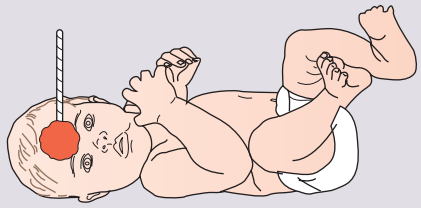
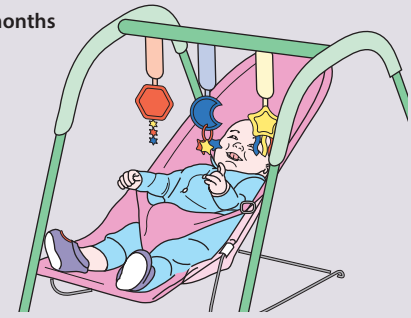
Vision and fine motor (median ages)

6 weeks



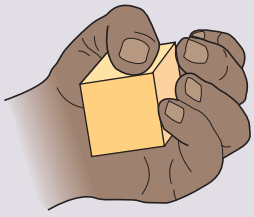
Follows moving object or face by turning the head (illustrated).

4 months



Reaches out for toys

4–6 months



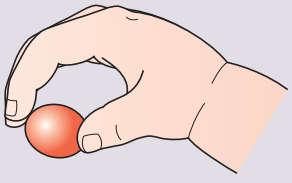
Palmar grasp

7 months



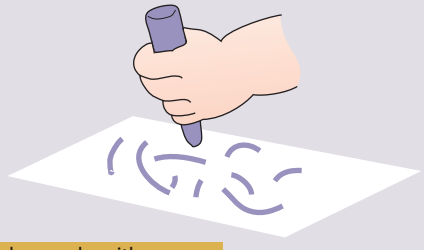
Transfers toys from one hand to another

10 months



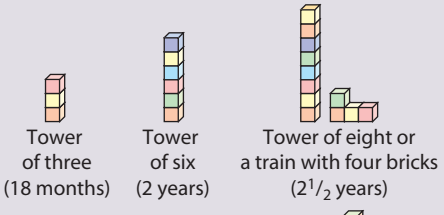
Mature pincer grip

16–18 months

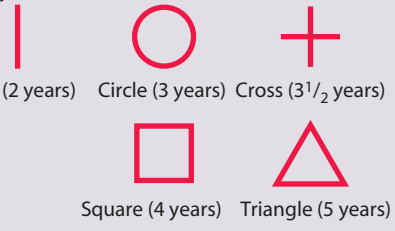


Makes marks with a crayon

14 months–4 years



2–5 years



Ability to draw without seeing how it is done. Can copy (draw after seeing it done) 6 months earlier.

Figure 3.5 Vision and fine motor skills (median ages).

Hearing, speech and language (median ages)

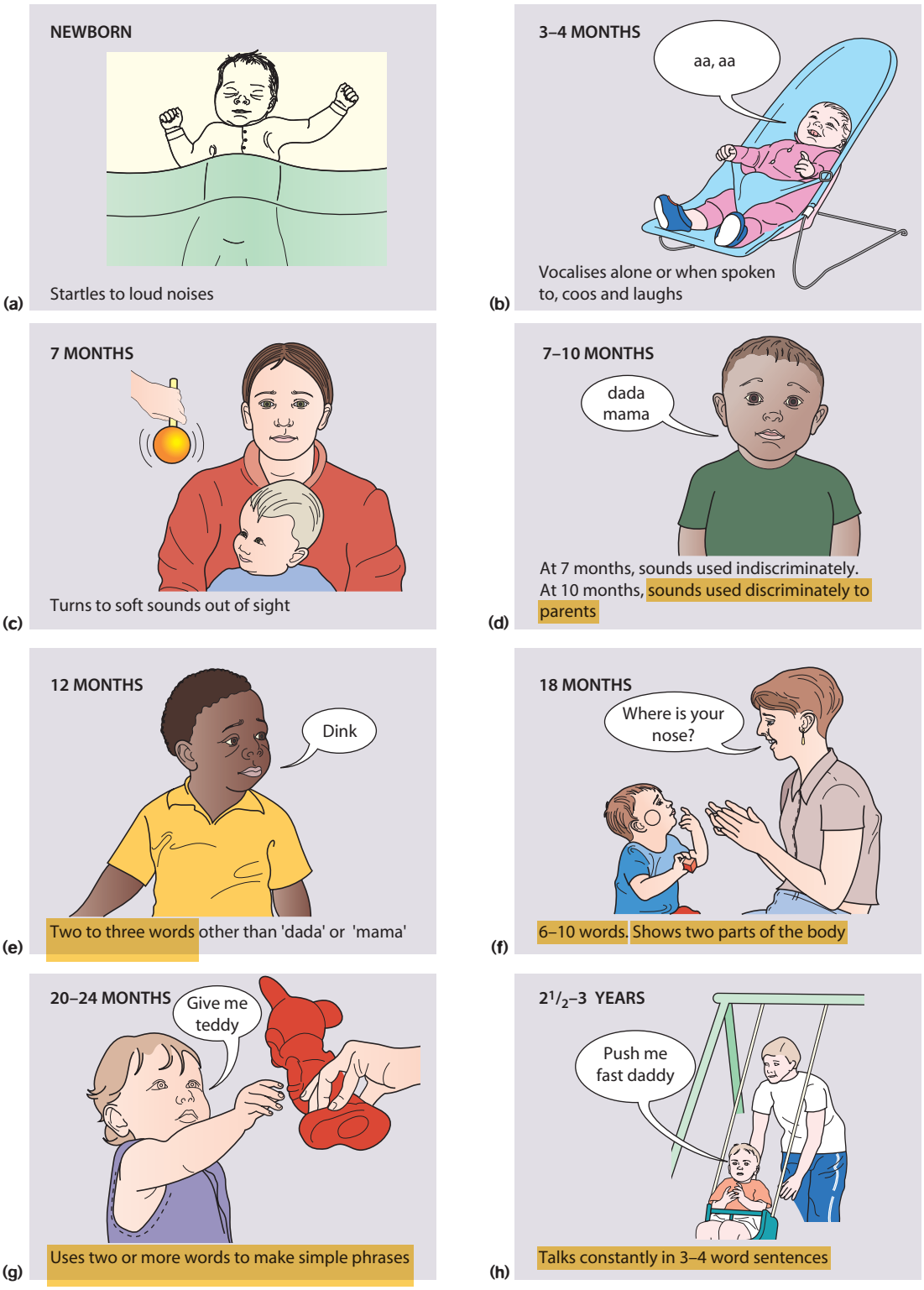


Figure 3.6 Hearing, speech and language (median ages).

Social, emotional and behavioural development (median ages)



Figure 3.7 Social, emotional and behavioural development (median ages).

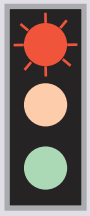
Summary

Fields of development with limit ages

Gross motor development

- Acquisition of tone and head control
- Primitive reflexes disappear
- Sitting
- Locomotor patterns
- Standing, walking, running
- Hopping, jumping, peddling

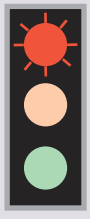
Gross motor	Limit ages
Head control	4 months
Sits unsupported	9 months
Stands independently	12 months
Walks independently	18 months



Vision and fine motor development

- Visual alertness, fixing and following
- Grasp reflex, hand regard
- Voluntary grasping, pincer, points
- Handles objects with both hands, transfers from hand to hand
- Writing, cutting, dressing

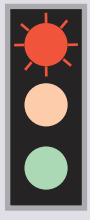
Vision and fine motor	Limit ages
Fixes and follows visually	3 months
Reaches for objects	6 months
Transfers	9 months
Pincer grip	12 months



Hearing, speech and language development

- Sound recognition, vocalisation
- Babbling
- Single words, understands simple requests
- Joining words, phrases
- Simple and complex conversation

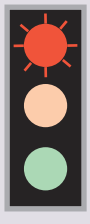
Hearing, speech and language	Limit ages
Polysyllabic babble	7 months
Consonant babble	10 months
Saying 6 words with meaning	18 months
Joins words	2 years
3-word sentences	2.5 years



Social, emotional, behaviour development

- Smiling, socially responsive
- Separation anxiety
- Self-help skills, feeding, dressing, toileting
- Peer group relationships
- Symbolic play
- Social/communication behaviour

Social behaviour	Limit ages
Smiles	8 weeks
Fear of strangers	10 months
Feeds self/spoon	18 months
Symbolic play	2–2.5 years
Interactive play	3–3.5 years



Summary

Developmental milestones by median age

Age	Gross motor	Vision and fine motor	Hearing, speech and language	Social, emotional and behavioural
Newborn	Flexed posture	Fixes and follows face	Stills to voice Startles to loud noise	Smiles – by 6 weeks
7 months	Sits without support	Transfers objects from hand to hand	Turns to voice Polysyllabic babble	Finger feeds Fears strangers
1 year	Stands independently	Pincer grip (10 months) Points	1–2 words Understands name	Drinks from cup Waves
15-18 months	Walks independently	Immature grip of pencil Random scribble	6–10 words Points to four body parts	Feeds self with spoon Beginning to help with dressing
2½ years	Runs and jumps	Draws	3–4 word sentences Understands two joined commands	Parallel play Clean and dry

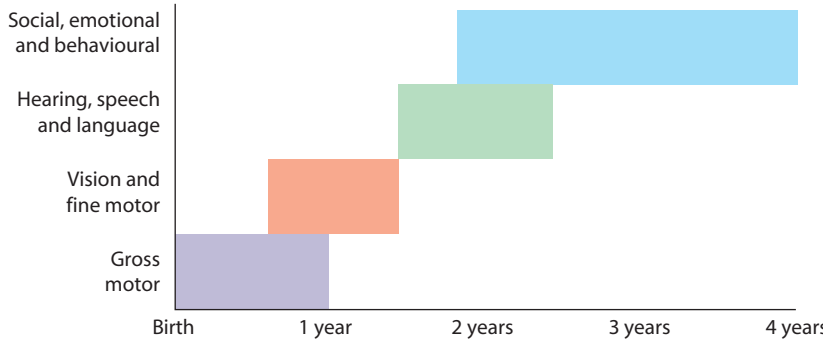


Figure 3.8 Diagram highlighting the ages when there is the most rapid emergence of skills in each developmental field.

- vision and fine motor development: more evident acquisition of skills from 1 year onwards
- hearing, speech and language: a big expansion of skills from 18 months
- social, emotional and behavioural development: expansion in skills is most obvious from 2.5 years.

Understanding the time when acceleration in each skill field becomes more obvious and knowing the child's age helps guide the direction of initial developmental questioning. Thus for a child aged:

- <18 months – it is likely to be most useful to begin questions around gross motor abilities, acquisition of vision and hearing skills, followed by questions about hand skills.
- 18 months to 2.5 years – initial developmental questioning is likely to be most usefully directed at acquisition of speech and language and fine motor (hand) skills with only later and brief questioning about gross motor skills (as it is likely the child would have presented earlier if these were of concern).
- 2.5 to 3.5 years – initial questions are best focused around speech and language and social, emotional and behaviour development.

Developmental questioning needs to cover all areas of developmental progression but this more focused way of taking a developmental history allows a useful short-cut approach. It directs the assessment to current abilities instead of concentrating on parents trying to remember the age when their child acquired developmental milestones some time in the past.

Observation during questioning

Of equal importance to taking the developmental history is the examiner's ability to observe the child throughout any visit. Not only will this provide an

almost immediate guide to where to begin questioning but also it will offer the opportunity for a rapid overview of the child's abilities, behaviour, peer group and parent–child relationships, all of which will go towards determining the overall picture about the child and his developmental abilities.

Equipment for developmental testing

Simple basic equipment is all that is needed for most developmental assessments. Equipment is aimed at bringing out the child's skills using play. Cubes, a ball, picture book, doll and miniature toys such as a tea-set, crayons and paper allow a quick but useful screen of mobility, hand skills, play, speech and language. These items allow the child to relax by having fun at the same time as facilitating observer assessment of his skills.

Developmental screening and assessment

Developmental screening (checks of whole populations of children at set ages by trained professionals) is a formal process within the child health surveillance and promotion programme. It is also an essential role of all health professionals to screen a young child's developmental progress opportunistically at every health contact, e.g. by the general practitioner for a sore throat, in the Accident and Emergency department for a fall or on admission to a paediatric ward. In this way, every child contact is optimised to check that development is progressing normally.

There are a number of problems inherent in developmental screening:

- It is a subjective clinical opinion, and therefore has its limitations
- A single observation of development may be limited by the child being tired, hungry, shy or simply not wishing to take part
- While much of the focus of early developmental progress in infants is centred on motor development, this is a poor predictor of problems in cognitive function and later school performance. Development of speech and language is a better predictor of cognitive function but is less easy to assess rapidly and may come at a time of less contact with health professionals.

The reliability of screening tests can be improved by adding a questionnaire completed by parents beforehand. Screening is being increasingly targeted towards children at high risk or when there are parental concerns.

Developmental assessment is the detailed analysis of particular areas of development and follows concern after screening that a child's developmental progress may be abnormal in some way. It is part of the diagnostic process and includes investigation, therapy and advice on how to optimise the child's progress. Developmental assessment is by referral to a specialist service and this may be the developmental paediatrician, therapy disciplines, or the local multidisciplinary child development service, which will include a paediatrician.

A range of tests have been developed to screen or to assess development in a formal reproducible manner. Screening tests include the Schedule of Growing Skills and the Denver Developmental Screening Test. Standardised tests that assess the development of infants and young children include the Griffiths and the Bailey Infant Development Scales. They are used, for example, in follow-up studies of preterm infants. There are also standardised tests concentrating on assessing specific aspects of development (e.g. the Reynell language scale, the Gross Motor Function Measure (GMFM) and the Autism Diagnostic Interview). All but the screening tests are time-consuming and require training for reliable results.

Cognitive function (higher mental function) can be assessed objectively with formal IQ tests by clinical or educational psychologists. However, IQ tests:

- may be affected by cultural background and linguistic skills
- do not test all skill areas
- do not necessarily reflect an individual child's ultimate potential
- may be compromised by individual disabilities, such as a motor disorder as in cerebral palsy.

'Verbal' intelligence tests, especially those for younger children, reflect general intellectual skills, particularly relating to language. 'Performance' or 'non-verbal' intelligence tests assess abilities independent of language. Verbal and performance intelligence testing allows formulation of a verbal IQ (VIQ) and performance IQ (PIQ), which together give an overall IQ figure.

Summary

Pattern of child development

When analysing a young child's developmental progress:

- Consider the child's age and then focus your questions on the areas of likely current developmental progress
- Offer the child suitable toys to find out about skills through play
- Observe how the child uses the toys and interacts with people.

Summary

Developmental screening and assessment

- *Developmental screening* – checks of whole populations or groups of children at set ages by trained professionals
- *Developmental assessment* – detailed analysis of overall development or specific areas of development.

Children with disabilities may have problems with speech or hand skills that may compromise testing, so that results in these situations have to be interpreted with care.

Cognitive (higher mental function) assessment of school-age children using IQ and other tests is carried out by clinical or educational psychologists.

Child health surveillance

In the UK, the healthy child programme (HCP) was introduced in 2009 (previously the child health promotion programme). It spans from pregnancy to 19 years old, but the main emphasis is on ages 0–5 years.

It offers families a programme of:

- Screening tests – early detection
- Immunisation – disease prevention
- Developmental reviews – including a specific screening at 2 years, an age when less obvious areas of developmental concern may arise with language skills
- Health promotion – information and guidance to support parenting and healthy choices.

There is a 'universal' programme, and a 'progressive' programme for families thought to be more at risk. Those in the progressive programme include infants or children with health or developmental problems, children at increased risk of obesity or families considered to be at higher risk, e.g. at-risk first-time mothers; parents with learning difficulties, drug or alcohol abuse or serious mental illness, insensitive (i.e. intrusive or passive) parenting interactions or domestic violence. These families receive additional intervention according to need. The programme is a compromise between the desire to detect problems and provide early intervention for all, while avoiding an excessive number of assessments. The way the healthy child programme (HCP) is organised is shown in Table 3.2. At each review, a check is made for specific physical abnormalities and on the child's overall development, health and growth. Selected health promotion topics are considered. There is an emphasis on parental opinion for vision, hearing, speech and language, as parents are usually excellent at the early detection of these problems. Details of each review are entered into the child's personal child health record kept by parents and brought whenever the child is seen by a health professional.

The HCP is carried out in primary care, usually by health visitors. If problems are identified, an action plan is made for the child, which could involve advice and monitoring progress or referral to the general practitioner or specialist.

Summary

The child health surveillance and promotion programme

- is provided in primary care
- includes immunisation, health promotion and developmental screening
- emphasises the role of parents in the early detection of developmental problems.

Hearing

During the later stages of pregnancy, the fetus responds to sound. At birth, a baby startles to sound, but there is a marked preference for voices. The ability to locate and turn towards sounds comes later in the first year. A checklist for parents of normal hearing responses during infancy is shown in Box 3.1.

Box 3.1 Hearing checklist for parents

Shortly after birth	Startles and blinks at a sudden noise, e.g. slamming of door
By 1 month	Notices sudden prolonged sounds, e.g. a vacuum cleaner, and pauses and listens when they begin
By 4 months	Quietens or smiles to the sound of your voice even when he cannot see you. He may also turn his head or eyes towards you if you come up from behind and speak to him from the side
By 7 months	Turns immediately to your voice across the room or to very quiet noises made on each side, so long as he is not too occupied with other things
By 9 months	Listens attentively to familiar everyday sounds and searches for very quiet sounds made out of sight. Should also show pleasure in babbling loudly and tunefully
By 12 months	Shows some response to his own name and to other familiar words. May respond when you say 'no' and 'bye-bye', even when he cannot see any accompanying gesture

If you suspect that your baby is not hearing normally, seek advice from your health visitor or doctor.

Used with permission from Dr Barry McCormick, Children's Hearing Assessment Centre, Nottingham, UK.

Table 3.2 The healthy child programme (HCP) provided by integrated local services, usually led by a health visitor

Age and checked by whom	Screening	General examination and immunisation	Health promotion
Antenatal (by 12th week of pregnancy) Midwife, GP or obstetrician	Antenatal screening for fetal conditions (see NICE guidelines for antenatal care)	Maternal health, fetal growth, maternal immunisation (rubella)	Universal: smoking and alcohol intake, mental health, breast-feeding
Newborn–72 h Usually hospital doctor; may be trained midwife, neonatal nurse practitioner or GP	Screening on examination includes eyes (red reflex), developmental dysplasia of the hip (DDH), testes in boys. Newborn Hearing Screening Programme Bilirubin check by 48 h if jaundiced	Normal newborn examination: general examination, weight and head circumference plotted on centile graph, BCG offered (repeated at 1, 2, 12 months) if at risk Hepatitis B vaccine if indicated (mother HepBsAg positive) Vitamin K	Universal: feeding, personal child health record and *Birth to Five* book, promoting sensitive parenting, prevention of sudden infant death syndrome
5–8 days (ideally Day 5) Midwife	Blood spot test for biochemical and haematological screening		
New baby review (by 14 days) Home visit by midwife or health visitor	Assess child and family health needs, including parental mental health needs	Examine baby for nutritional status and prolonged jaundice	Infant feeding, promote sensitive parenting, advice on promoting development, home safety
6–8 weeks General practitioner	Physical examination: cardiac abnormalities (heart murmurs and femoral pulses), DDH, testicular descent in boys, red reflex of fundus, matters of parental concern	Full physical examination, weight, head circumference and plot growth centiles Vision/hearing – any parental concern? 1st immunisation – DTaP/IPV, Hib, PCV	Nutrition, immunisations, recognition of illness, avoid passive smoking, crying and sleep problems, maternal mental health
3 months Child health clinic	General review of progress, address parental concerns such as growth	2nd immunisation – DTaP/IPV/Hib, MenC	Support families by providing access to parenting and child health information
4 months Child health clinic	General review of progress	Third immunisation – DTaP/IPV/Hib, PCV MenC	Weaning on to solids around 6 months
7–9 months Health visiting team	Systematic assessment of the child's physical, emotional and social development and family needs	If parental concern – hearing, vision, development, growth	Distribution of books,[a] accident prevention: choking, scalds and burns, safety gates, nutrition and dental care, skin care (sunburn)
12–13 months Child health clinic	General review of progress	Immunisation – Hib, MenC, PCV, MMR	Dental health
2–2½ year review Health visiting team coordinate this key review	Nutrition, active play, personal, social and emotional development, speech, language and communication	Review immunisation status and physical status according to parental concerns	Obesity prevention, injury prevention, advice on how to seek medical help
3–5 years (preschool) Heath visiting team	General review of progress	3-4 years immunisation – MMR, DTaP/IPV	Health promotion and supporting parents
By 5 years (to be completed soon after school entry) School nurse	Orthoptist: screen all children for visual impairment (4–5 years) School nurse: hearing screening (audiometry), growth	Review immunisation status Measure height and weight, plot centiles[b] Physical examination if parental concern	Health promotion and supporting parents
5–11 years School nurse	Share information about preschool background, assessment of emotional health, psychological well-being and mental health	Nursing care provided according to needs	Promote healthy weight, support for parents and carers
11–16 years School nurse	Health review at school transition at 10–11 and 15–16 years by questionnaires, engaging primary care in mid-teens, emotional health, psychological well-being and mental health	Immunisation (13–16 years) – Td/IPV Human papilloma virus (HPV) in girls If at risk – BCG, Hep B, influenza	Sexual health, promote healthy weight
16–19 years (Further Education) School nurse, GP	Share information from school with adult services, emotional health, psychological well-being and mental health	Review immunisation status	Sexual health, encourage physical activity, support for parents and carers

Adapted from: The Healthy Child Programme 2009, Department of Health, London, UK.
DTaP/IPV: Diphtheria, tetanus, pertussis, polio immunization; Hib: *Haemophilus B influenzae*; PCV: Pneumococcal; MenC: Meningitis C; Td/IPV: Diphtheria, tetanus, polio; MMR: Measles, Mumps and Rubella
[a]Bookstart – national programme to encourage parents and carers to enjoy books with their children from an early age.
[b]The national child measurement programme – height and weight of all reception and year 6 children.

Hearing tests

Newborn

Early detection and treatment of hearing impairment improves the outcome for speech and language and behaviour. In order to detect hearing impairment in the newborn period, hearing can be tested by:

- Evoked otoacoustic emission (EOAE) (Fig. 3.9a) – an earphone produces a sound which evokes an echo or emission from the ear if cochlear function is normal.
- Auditory brainstem response (ABR) audiometry (Fig. 3.9b) – computer analysis of EEG waveforms evoked in response to a series of auditory stimuli.

Universal neonatal hearing screening has been introduced in many countries. In the UK, initial screening is performed using different combinations of EOAE testing or ABR audiometry. If a normal response cannot be obtained, the child is referred to an audiologist.

Distraction testing

This was the mainstay of hearing screening but has been replaced by universal neonatal screening. It is now only used as a screening test for infants who have not had newborn screening or as a diagnostic test. It is performed at 7–9 months of age (Fig. 3.10). The test relies on the baby locating and turning appropriately towards sounds. High- and low-frequency sounds are presented out of the infant's field of vision. Testing is unreliable if not carried out by properly trained staff, since it can be difficult to identify hearing-impaired infants as they are particularly adept at using non-auditory cues.

Visual reinforcement audiometry

This is particularly useful to assess impairment in infants between 10 and 18 months, although it can be used between the age of 6 months and 3 years. Hearing thresholds are established using visual rewards (illumination of toys) to reinforce the child's head turn to stimuli of different frequencies. Localisation of the stimuli is not necessary and insert earphones may be used to obtain ear-specific information, thus making it more useful than free field tests such as distraction and performance testing (Fig. 3.11).

Performance and speech discrimination testing

Performance testing using high- and low-frequency stimuli and speech discrimination testing using miniature toys can be used for children with suspected hearing loss at 18 months to 4 years of age (Fig. 3.12).

Audiometry

Threshold audiometry using headphones, where the child responds to a pure tone stimulus, can be used to detect and assess the severity of hearing loss in children from 4 years old.

Parental concern

At all ages, parental concern about hearing warrants further assessment.

Summary

Hearing

- Early detection and treatment of hearing impairment improves the outcome of speech and language, and behaviour.
- Newborn hearing screening is performed for the early identification of hearing impairment.
- If there is parental concern about hearing, further assessment is warranted.

Vision

A newborn infant's vision is limited; the visual acuity is only about 6/200 (can see at 6 metres what a normally sighted adult can see at 200 metres). The peripheral retina is well developed but the fovea is immature and the optic nerve unmyelinated. Well-focused images on the retina are required for the acquisition of visual acuity and any obstruction to this, e.g. from a cataract, will interfere with the normal development of the optic pathways and visual cortex unless corrected early in life.

Many newborn infants can fix and follow horizontally a face or coloured ball or the image of a target of concentric black and white circles. Most squint transiently, particularly when the baby tries to look at near objects and the eyes over-converge.

By about 6 weeks of age, both eyes should move together when following a light source. By 12 weeks, no squint should be present. Babies slowly develop the ability to focus at different distances. Visual acuity also improves: from 6/60 at 3 months to being able to poke at 1 cm objects at 8 months and at 1 mm objects (e.g. 'hundreds and thousands' cake sprinkles) at 15 months. Adult levels are reached by 3–4 years of age, when the child can match pictures or letters at 6/6 using both eyes together.

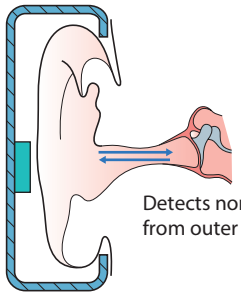
Summary

Vision

- Term newborn infants can fix and follow horizontally and prefer to watch faces.
- Visual acuity is poor in the newborn but increases to adult levels by 4 years of age.
- Vision screening is performed at school entry or in preschool children.

Hearing screening of newborn infants

(a) Evoked otoacoustic emission (EOAE)



Click generated from ear phones

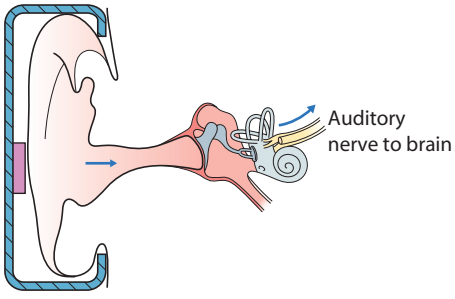
Detects normal sound vibrations from outer hair cells in the cochlea

Advantages:
- Simple and quick to perform, though is affected by ambient noise

Disadvantages:
- Misses auditory neuropathy as function of auditory nerve or brain not tested
- Relatively high false-positive rate in first 24 hours after birth as vernix or amniotic fluid are still in ear canal
- Not a test of hearing but a test of cochlear function

(b) Automated auditory brainstem response (AABR)



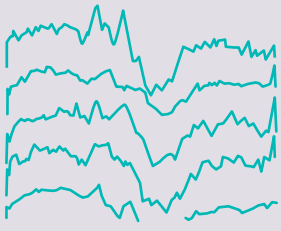


Auditory nerve to brain

Auditory stimulus via earphones

Signal via ear and auditory nerve to brain

EEG waveforms – computerised analysis determines if normal or abnormal



Advantages:
- Screens hearing pathway from ear to brainstem
- Low false-positive rate

Disadvantages:
- Affected by movement, so infants need to be asleep or very quiet, so time consuming
- Complex computerised equipment, but is mobile
- Requires electrodes applied to infant's head, which parents may dislike

Figure 3.9 Universal neonatal hearing screening is usually performed using **(a)** otoacoustic emission testing or **(b)** auditory brainstem response audiometry.

Hearing tests

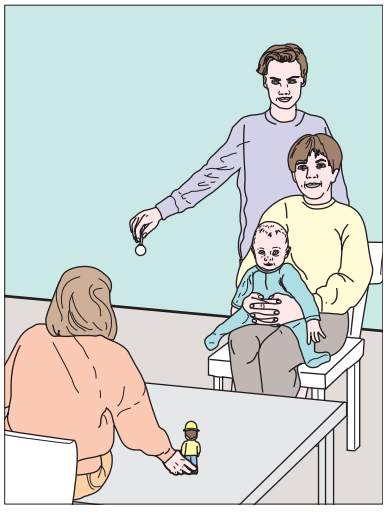


Figure 3.10 Distraction hearing test. The test is hard to perform reliably as babies with hearing difficulties learn to compensate by using shadows, smells and guesswork to locate the presenter. The test must be done by well-trained professionals.

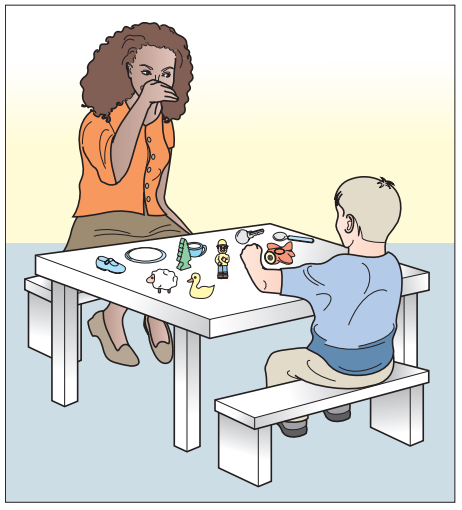


Figure 3.12 Speech discrimination testing using miniature toys to detect hearing loss in children between 18 months and 4 years of age.

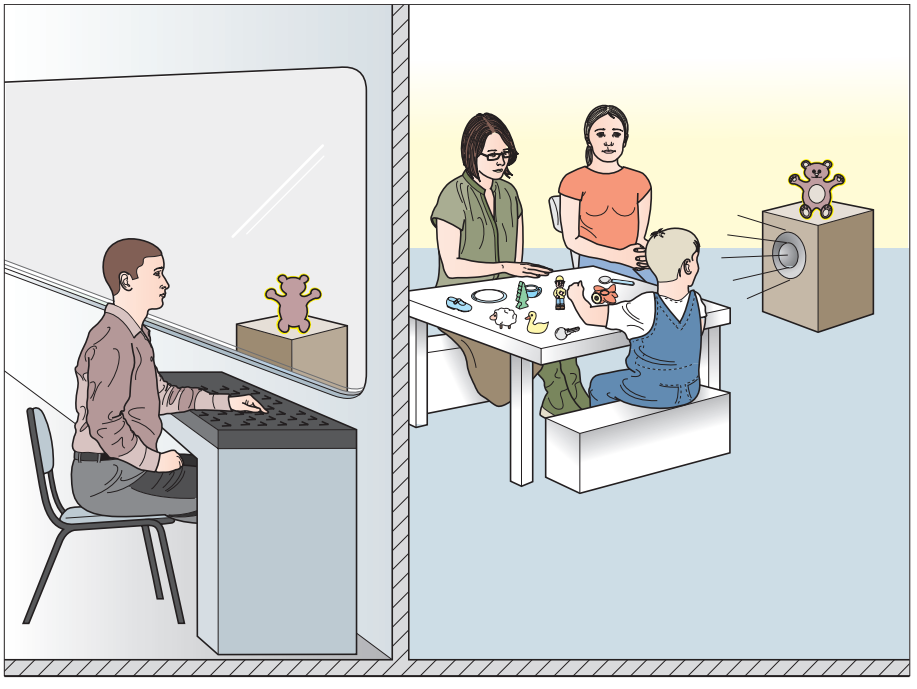


Figure 3.11 Visual reinforcement audiometry. While an assistant plays with the child, sounds of a specific frequency are emitted from a speaker. When the child turns to it, the tester lights up a toy by the speaker to reinforce the sound with a visual reward. This test is particularly useful at 10–18 months.

Vision testing

The assessment of vision at different ages is shown in Table 3.3. All children in the UK are screened for visual acuity and squint at school entry. In some parts of the UK, screening is carried out in preschool children at 4–5 years.

Table 3.3 Testing vision at different ages

Age	Test
Birth	Face fixation and following
6–8 weeks	Fix and follow bright toy, optokinetic nystagmus
6 months	Reaches well for toys, preferential looking tests
2½ years	Can identify or match pictures of reducing size (Kay pictures)
4 years	Can identify or match letters in linear optotype book (Crowded LogMAR)
6 years onwards	Can identify or match letters on a LogMAR chart

Note: Using single letters/pictures instead of lines underestimates amblyopia severity (see Ch. 4).

Further reading

Meggitt C: *Child Development*, Oxford, 2007, Heinemann.

Website (accessed May 2011)

Department of Health: *Birth to Five*, 2009. Available at www.nhs.uk/birthtofive.

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4

Developmental problems and the child with special needs

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Any child whose development is delayed or disordered needs assessment to determine the cause and management. Neurodevelopmental problems present at all ages, with an increasing number now recognised antenatally (Table 4.1). Many are identified in the neonatal period because of abnormal neurology or dysmorphic features. During infancy and early childhood, problems often present at an age when a specific area of development is most rapid and prominent, i.e. motor problems during the first 18 months of age, speech and language problems between 18 months and 3 years and social and communication disorders between 2 and 4 years. Abnormal development may be caused not only by neurodevelopmental problems (Table 4.2) but also by ill health or if the child's physical or psychological needs are not met.

When performing a clinical examination on a young child with a developmental problem:

- Ask the parent what their child can and cannot do. Start at a level below what a child is likely to be able to do to retain confidence of the parent and child.
- *Observe* the child from the first moment seen.
- Make it fun. Your examination should be perceived as a game by the child, although they may not always follow your rules.
- Toys to use are cubes, a ball, car, doll, pencil, paper, pegboard, miniature toys, picture book. Adapt their use to the child.
- Formulate a developmental picture in terms of gross motor; vision and fine motor; hearing, speech and language; and social, emotional and behaviour. As you become more confident you will screen all these skills simultaneously.
- At the end of developmental screening you should be able to describe what a child is able to do and what the child cannot do, if the abilities are within normal limits for age and, if not, which developmental fields are outside the normal range.
- Clinical signs to look for that may aid diagnosis or guide investigation are:
  - patterns of growth: height, weight, head circumference with centile plotting
  - dysmorphic features: face, limbs, body proportions, cardiac, genitalia
  - skin: neurocutaneous stigmata, injuries, cleanliness
  - central nervous system examination: abnormal posture/symmetry, wasting, power and tone, deep tendon reflexes, clonus, plantar responses, sensory examination, cranial nerves
  - cardiovascular examination: abnormalities are associated with many dysmorphic syndromes
  - visual function and ocular abnormalities
  - hearing: by questioning parents about hearing and language development and checking if neonatal hearing screening was done

Table 4.1 Features that may suggest neurodevelopmental concerns by age

Prenatal	Positive family history, e.g. affected siblings or family members; ethnicity, e.g. Tay–Sachs disease in Jewish parents
	Antenatal screening tests, e.g. ultrasound including nuchal thickness and triple blood test for conditions such as Down syndrome, neural tube defects (spina bifida) and hydrocephalus. Amniocentesis for chromosomal disorders
Perinatal	Following birth asphyxia/neonatal encephalopathy
	Preterm infants with intraventricular haemorrhage/periventricular leucomalacia, post-haemorrhagic hydrocephalus
	Dysmorphic features
	Abnormal neurological behaviour – tone, feeding, movement, seizures, visual inattention
Infancy	Global developmental delay
	Delayed or asymmetric motor development
	Vision or hearing concerns by parent or after screening
	Neurocutaneous/dysmorphic features
Preschool	Speech and language delay
	Abnormal gait, clumsy motor skills
	Poor social communication skills
School age	Problems with balance and coordination
	Learning difficulties
	Attention control
	Hyperactivity
	Specific learning difficulties, e.g. dyslexia, dyspraxia
	Social communication difficulties
Any age	Acquired brain injury, e.g. after meningitis, head injury
	Loss of skills

  - patterns of mobility, dexterity, hand dominance, communication and social skills, general behaviour
  - cognition.

Many examination findings can be predicted from *observation* of functional skills.

Many parental concerns about their child's development are found to be variations of the norm, in which case the parents should be reassured. If in doubt, observe the child's progress over a period of time.

Abnormal development – key concepts

The terminology can be confusing, but:

- *Delay* – implies slow acquisition of all skills (global delay) or of one particular field or area of skill (specific delay), particularly in relation to developmental problems in the 0–5 years age group
- *Learning difficulty* – used in relation to children of school age and may be cognitive, physical, both or relate to specific functional skills
- *Disorder* – maldevelopment of a skill.

The following are agreed definitions:

- *Impairment* – loss or abnormality of physiological function or anatomical structure
- *Disability* – any restriction or lack of ability due to the impairment
- *Disadvantage* – this results from the disability, and limits or prevents fulfilment of a normal role. It is situationally specific; a child with a learning disability may for example be a good skier or enjoy swimming.

The term 'handicap' is now discouraged as it can imply a person deserves pity. Difficulty and disability are often used interchangeably, but difficulty is used particularly in an educational context.

The *pattern* of abnormal development (global or specific) can be categorised as (Fig. 4.1):

Table 4.2 Conditions which cause abnormal development and learning difficulty

Prenatal	
Genetic	Chromosome/DNA disorders, e.g. Down syndrome, fragile X syndrome, chromosome microdeletions or duplications
	Cerebral dysgenesis, e.g. microcephaly, absent corpus callosum, hydrocephalus, neuronal migration disorder
Vascular	Occlusions, haemorrhage
Metabolic	Hypothyroidism, phenylketonuria
Teratogenic	Alcohol and drug abuse
Congenital infection	Rubella, cytomegalovirus, toxoplasmosis, HIV
Neurocutaneous syndromes	Tuberous sclerosis, neurofibromatosis
Perinatal	
Extreme prematurity	Intraventricular haemorrhage/periventricular leucomalacia
Birth asphyxia	Hypoxic-ischaemic encephalopathy
Metabolic	Symptomatic hypoglycaemia, hyperbilirubinaemia
Postnatal	
Infection	Meningitis, encephalitis
Anoxia	Suffocation, near drowning, seizures
Trauma	Head injury – accidental or non-accidental
Metabolic	Hypoglycaemia, inborn errors of metabolism
Vascular	Stroke
Other	
Unknown (about 25%)	

The site and severity of brain damage influences the clinical outcome, i.e. whether specific or global developmental delay, learning and/or physical disability.

- slow but steady
- plateau effect
- showing regression.

The *severity* can be categorised as:

- mild
- moderate
- severe
- profound.

Other features of developmental delay are:

- The gap between normal and abnormal development becomes greater with increasing age and therefore becomes more apparent over time (Fig. 4.2).
- It may be the presentation of a wide variety of underlying conditions (Table 4.2).
- The site and severity of brain damage influences the clinical outcome, i.e. whether there will be specific or global developmental delay, learning and/or physical disability.
- It may be genetic, with important implications for the family.
- There is a wide age band across which it can be normal to achieve a developmental skill. Limit ages denote beyond the normal range.

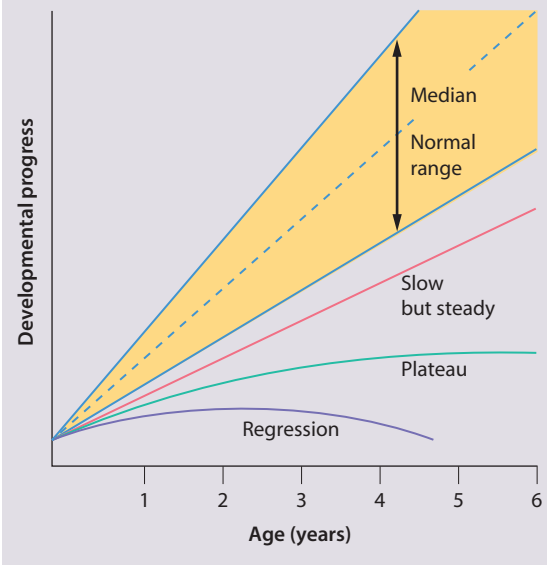


Figure 4.1 Patterns of abnormal development. These may be slow but steady, plateau or regression.

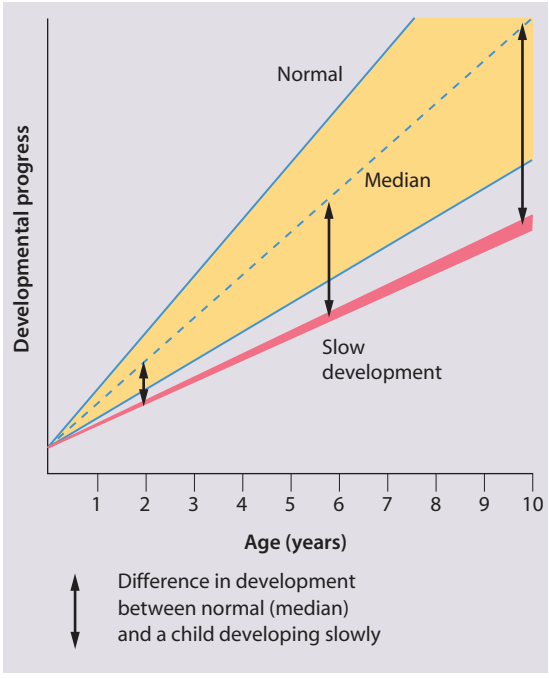


Figure 4.2 For children with abnormal development, the gap between their abilities and what is normal widens with age.

The choice of investigations to identify the cause is influenced by the child's age, the history and clinical findings (Table 4.3). In some children, no cause can be identified even after extensive investigation.

Summary

Abnormal development

- Incorporates global and specific delay or disorder, learning difficulty, impairment and disability
- Varies in pattern of progression and severity
- Becomes more apparent with age.

Developmental delay

Global developmental delay (also called early developmental impairment) implies delay in acquisition of all skill fields (gross motor, vision and fine motor, hearing and speech, and language and cognition, social/emotional and behaviour). It usually becomes apparent in the first 2 years of life. Global developmental delay is likely to be associated with cognitive difficulties, although these may only become apparent several years later. The presence of global developmental delay should always generate investigation into a possible cause such as those listed in Table 4.2. When children become older and the clinical picture is clearer, it is more appropriate to describe the individual difficulties such as learning disability, motor disorder and communication difficulty, rather than using the term global developmental delay.

Specific developmental impairment is when one field of development or skill area is more delayed than others. It may also be developing in a disordered way.

Global developmental delay usually presents in the first 2 years of life.

Abnormal motor development

This may present as delay in acquisition of motor skills, e.g. head control, rolling, sitting, standing, walking or as problems with balance, an abnormal gait, asymmetry of hand use, involuntary movements or rarely loss of motor skills. Concern about motor development usually presents between 3 months and 2 years of age when acquisition of motor skills is occurring most rapidly. Examination may reveal underlying abnormal motor signs.

Causes of abnormal motor development include:

- central motor deficit e.g. cerebral palsy
- congenital myopathy/primary muscle disease
- spinal cord lesions, e.g. spina bifida
- global developmental delay, as in many syndromes or of unidentified cause.

As hand dominance is not acquired until 1–2 years or later, asymmetry of motor skills during the first year of life is always abnormal and may suggest an underlying hemiplegia.

Late walking (>18 months old) may be caused by any of the above but also needs to be differentiated from children who display the normal locomotor variants of bottom-shuffling or commando crawling (see Ch. 3), where walking occurs later than with crawlers.

Concern about abnormal motor development needs assessment by a neurodevelopmental paediatrician and physiotherapist. Ongoing physiotherapy input and subsequent involvement of an occupational therapist is likely to be needed.

Cerebral palsy (CP)

Cerebral palsy may be defined as an abnormality of movement and posture, causing activity limitation attributed to non-progressive disturbances that occurred in the developing fetal or infant brain. The motor disorders of CP are often accompanied by disturbances of cognition, communication, perception, sensation, behaviour and seizure disorder and secondary musculoskeletal problems. Although the lesion is non-progressive, the clinical manifestations emerge over time, reflecting the balance between normal and abnormal cerebral maturation. Cerebral palsy is the most common cause of motor impairment in children, affecting about 2 per 1000 live births. The term is usually used for brain injuries occurring up to the age of 2 years. After this age, it is more appropriate to use acquired brain injury as the diagnosis. Although the

Table 4.3 Investigations or assessment to consider for developmental delay

Cytogenetic	Chromosome karyotype[a]
	Fragile X analysis[a]
	DNA FISH analysis, e.g. for chromosome 7, 15, 22 deletions, CGH microarray (comparative genomic hybridisation), telomere screen
Metabolic	Thyroid function tests, liver function tests, bone chemistry, urea and electrolytes, plasma amino acids[a]
	Creatine kinase, blood lactate, VLCFA (very long chain fatty acids), ammonia, blood gases, white cell (lysosomal) enzymes, urine amino and organic acids, urine mucopolysaccharides (GAG) and oligosaccharide screen, urine reducing substances, lead levels, urate, ferritin, biotinidase
	Maternal amino acids for raised phenylalanine
Infection	Congenital infection screen
Imaging	Cranial ultrasound in newborn
	CT and MRI brain scans
	Skeletal survey, bone age
Neurophysiology	EEG (for seizures and can be specific for some progressive neurological disorders and syndromes)
	Nerve conduction studies, EMG, VEP (visual evoked potentials), ERG (electroretinogram)
Histopathology/ histochemistry	Nerve and muscle biopsy
Other	Hearing[a]
	Vision[a]
	Clinical genetics
	Cognitive assessment
	Therapy assessment – physiotherapy, occupational therapy and speech and language therapy
	Child psychiatry
	Dietician
	Nursery/school reports

[a]Basic screening tests.

underlying cause is static, the resulting motor disorder may evolve, giving the impression of deterioration. The diagnosis for each child should formulate: the distribution of the motor disorder, the movement type, the cause and any associated impairment.

Causes

About 80% of cerebral palsy is antenatal in origin due to vascular occlusion, cortical migration disorders or structural maldevelopment of the brain during gestation. Some of these problems are linked to gene deletions. Other antenatal causes are genetic syndromes and congenital infection.

Only about 10% of cases are thought to be due to hypoxic-ischaemic injury during delivery and this proportion has remained relatively constant over the last decade. About 10% are postnatal in origin.

Preterm infants are especially vulnerable to brain damage from periventricular leucomalacia (PVL) secondary to ischaemia and/or severe intraventricular haemorrhage. The rise in survival of extremely preterm infants has been accompanied by an increase in survivors with cerebral palsy, although the number of such children is relatively small.

Postnatal causes are meningitis/encephalitis/encephalopathy, head trauma from accidental or non-accidental injury, symptomatic hypoglycaemia, hydrocephalus and hyperbilirubinaemia.

MRI brain scans may assist in identifying the cause of the cerebral palsy but a scan is not necessary to make the diagnosis.

Clinical presentation

Many children who develop cerebral palsy will have been identified as being at risk in the neonatal period. Early features of cerebral palsy are as follows:

- Abnormal limb and/or trunk posture and tone in infancy with delayed motor milestones (Fig. 4.3); may be accompanied by slowing of head growth
- Feeding difficulties, with oromotor incoordination, slow feeding, gagging and vomiting
- Abnormal gait once walking is achieved
- Asymmetric hand function before 12 months of age.

In CP, primitive reflexes, which facilitate the emergence of normal patterns of movement and which need to disappear for motor development to progress, may persist and become obligatory (see Ch. 3).

The diagnosis is made by clinical examination, with particular attention to assessment of posture and the pattern of tone in the limbs and trunk, hand function and gait. There are three main clinical subtypes: spastic (90%), dyskinetic (6%) and ataxic (4%). A mixed pattern may occur. Functional ability is described using the Gross Motor Function Classification System (Table 4.4).

Spastic cerebral palsy

In this type, there is damage to the upper motor neurone (pyramidal or corticospinal tract) pathway. Limb tone is persistently increased (spasticity) with associated brisk deep tendon reflexes and extensor plantar responses. The tone in spasticity is velocity dependent, so the faster the muscle is stretched the greater the resistance it will have. This elicits a dynamic catch which is the hallmark of spasticity. The increased limb tone may suddenly yield under pressure in a 'clasp knife' fashion. Limb involvement is increasingly described as unilateral or bilateral to acknowledge asymmetrical signs. Spasticity tends to present early and may even be seen in the neonatal period. Sometimes there is initial hypotonia, particularly of the head and trunk. There are three main types of spastic cerebral palsy:

- *Hemiplegia* – unilateral involvement of the arm and leg (Fig. 4.4). The arm is usually affected more than the leg, with the face spared. Affected children often present at 4–12 months of age with fisting of the affected hand, a flexed arm, a pronated forearm, asymmetric reaching or hand function. Subsequently a tiptoe walk (toe–heel gait) on the affected side may become evident. Affected limbs may initially be flaccid and hypotonic, but increased tone soon emerges as the predominant sign. The past medical history may be normal, with an unremarkable birth history with no evidence of hypoxic-ischaemic encephalopathy. In some, the condition is caused by neonatal stroke. Larger brain lesions (strokes) may cause a hemianopia (loss of half of visual field) of the same side as the affected limbs
- *Quadriplegia* – all four limbs are affected, often severely. The trunk is involved with a tendency to opisothonus (extensor posturing), poor head control and low central tone (Fig. 4.5). This more severe cerebral palsy is often associated with seizures, microcephaly and moderate or severe intellectual impairment. There may have been a history of perinatal hypoxic-ischaemic encephalopathy.
- *Diplegia* – all four limbs, but the legs are affected to a much greater degree than the arms, so that hand function may appear to be relatively normal. Motor difficulties in the arms are most apparent with functional use of the hands. Walking is abnormal. Diplegia is one of the patterns associated with preterm birth due to periventricular brain damage.

Table 4.4 Gross Motor Function Classification System (GMFCS)

Level I	Walks without limitations
Level II	Walks with limitations
Level III	Walks using a handheld mobility device
Level IV	Self-mobility with limitations; may use powered mobility
Level V	Transported in a manual wheelchair

See http://www.canchild.ca/en/measures/gmfcs.asp for further details (accessed January 2011).

Dyskinetic cerebral palsy

Dyskinesia refers to movements which are involuntary, uncontrolled, occasionally stereotyped, and often more evident with active movement or stress. Muscle tone is variable and primitive motor reflex patterns predominate. May be described as:

- Chorea – irregular, sudden and brief non-repetitive movements
- Athetosis – slow writhing movements occurring more distally such as fanning of the fingers
- Dystonia – simultaneous contraction of agonist and antagonist muscles of the trunk and proximal muscles often giving a twisting appearance.

Intellect may be relatively unimpaired. Affected children often present with floppiness, poor trunk control and delayed motor development in infancy. Abnormal movements may only appear towards the end of the first year of life. The signs are due to damage or dysfunction in the basal ganglia or their associated pathways (extrapyramidal). In the past the commonest cause was hyperbilirubinaemia (kernicterus) due to rhesus disease of the newborn but it is now hypoxic-ischaemic encephalopathy at term.

Ataxic (hypotonic) cerebral palsy

Most are genetically determined. When due to acquired brain injury (cerebellum or its connections), the signs occur on the same side as the lesion but are usually relatively symmetrical. There is early trunk and limb hypotonia, poor balance and delayed motor development. Incoordinate movements, intention tremor and an ataxic gait may be evident later.

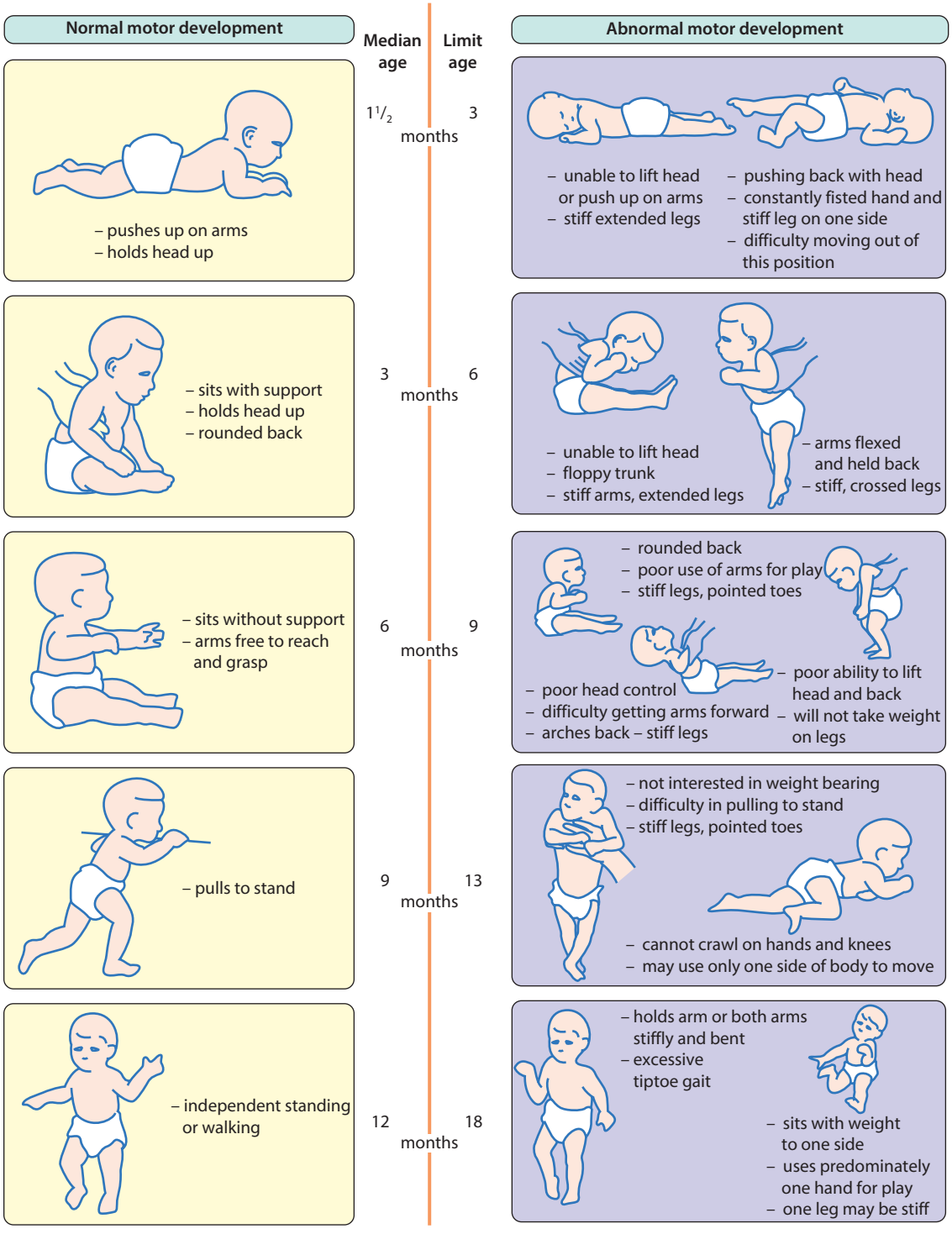


Figure 4.3 Normal motor milestones and patterns of abnormal motor development. Cerebral palsy (hemiplegia or quadriplegia) is the commonest cause of developmental problems. (Adapted from Pathways Awareness Foundation, 123 North Wacker Drive, Chicago, IL. Tel. (+1) 800 326 8154; see also http://www.pathwaysawareness.org. Accessed May 2011.)

Cerebral palsy



Figure 4.4 A child with a right spastic hemiplegia. His right arm is hyperpronated, flexed, hand fisted.



Figure 4.5 A child with spastic quadriplegia showing scissoring of the legs from excessive adduction of the hips, pronated forearms and 'fisted' hands.

Management

Parents should be given details of the diagnosis as early as possible, but prognosis is difficult during infancy until the severity and pattern of evolving signs and the child's developmental progress have become clearer over several months or years of life. Children with cerebral palsy are likely to have a wide range of associated medical, psychological and social problems, making it essential to adopt a multidisciplinary approach to assessment and management, as described later in this chapter.

Summary

Cerebral palsy

- has many causes. Only about 10% follow hypoxic-ischaemic encephalopathy
- usually presents in infancy with abnormal tone and posture, delayed motor milestones and feeding difficulties
- may be spastic, dyskinetic, ataxic or a mixed pattern.

Abnormal speech and language development

A child may have a deficit in either receptive or expressive speech and language, or both. The deficit may be a delay or a disorder.

Speech and language *delay* may be due to:

- hearing loss
- global developmental delay
- difficulty in speech production from an anatomical deficit, e.g. cleft palate, or oromotor incoordination, e.g. cerebral palsy
- environmental deprivation/lack of opportunity for social interaction
- normal variant/familial pattern.

Speech and language *disorders* include disorders of:

- language comprehension
- language expression – inability or difficulty in producing speech whilst knowing what is needing to be said
- phonation and speech production such as stammering (dysfluency), dysarthria or verbal dyspraxia
- pragmatics (difference between sentence meaning and speaker's meaning), construction of sentences, semantics, grammar
- social/communication skills (autistic spectrum disorder).

Speech and language problems are usually first suspected by parents or primary healthcare professionals. A hearing test and assessment by a speech and language therapist are the initial steps. In early years, there is considerable overlap between language and cognitive (intellectual) development. Involvement of a neurodevelopmental paediatrician and paediatric audiological physician is indicated. Speech and language therapy may be provided on a continuous, burst or review basis. The speech therapist may promote alternative methods of communication such as signing (with Makaton or the Picture Exchange Communication System, PECS). Special schooling (usually language units attached to a mainstream primary school) are available but only appropriate for a very few. Many children with early speech and language problems will need learning support at school entry.

There are many tests of language development. These include:

- The Symbolic Toy test, which assesses very early language development.
- The Reynell test for receptive and expressive language, used for preschool children.

Abnormal development of social/communication skills (autistic spectrum disorders)

Children who fail to acquire normal social and communication skills may have an autistic spectrum disorder. The prevalence of autistic spectrum disorder is 3–6/1000 live births. It is more common in boys. Presentation is usually between 2 and 4 years of age when language and social skills normally rapidly expand. The child presents with a triad of difficulties and associated co-morbidities (Box 4.1).

Where only some of the behaviours are present, the child may be described as having autistic features but not the full spectrum.

Asperger syndrome refers to a child with the social impairments of an autistic spectrum disorder but at the milder end, and near-normal speech development. Such children still have major difficulties with the give-and-take of ordinary social encounters, a stilted way of speaking and narrow, strange interests which they do not share with others, and are often clumsy. In reality autistic spectrum disorders are a continuum of behavioural states ranging from the severe form of autism with or without severe learning difficulties to the milder Asperger syndrome, to autistic features occurring secondary to other clinical problems. Autism is diagnosed by observation of behaviour, including the use of formal standardised tests. It may arise as the result of different organic processes but in many cases no specific cause can be identified. There is probably multiple aetiology with a genetic component in at least some. The condition is not the result of emotional trauma or deviant parenting. There is no evidence for a suggested link with the MMR vaccine.

Management

The condition has lifelong consequences of varying degree for the child's social/communication and learning skills. Parents need a great deal of support. They often feel initial guilt that they did not recognise the problem earlier. A wide range of interventions have been promoted over the last 10 years but with little evidence except for applied behavioural analysis (ABA), a behaviour modification approach that helps to reduce ritualistic behaviour, develop language, social skills and play and to generalise use of all these skills. It is currently the most widely accepted treatment approach but requires 25–30 h of individual therapy each week, so is costly and time-consuming. An appropriate educational placement needs to be sought; some schools incorporate this approach. Less than 10% of children with autism are able to function independently as adults.

Box 4.1 Features of autistic spectrum disorders

Impaired social interaction:

- does not seek comfort, share pleasure, form close friendships
- prefers own company, no interest or ability in interacting with peers (play or emotions)
- gaze avoidance
- lack of joint attention
- socially and emotionally inappropriate behaviour
- does not appreciate that others have thoughts and feelings
- lack of appreciation of social cues

Speech and language disorder:

- delayed development, may be severe
- limited use of gestures and facial expression
- formal pedantic language, monotonous voice
- impaired comprehension with over-literal interpretation of speech
- echoes questions, repeats instructions, refers to self as 'you'
- can have superficially good expressive speech

Imposition of routines with ritualistic and repetitive behaviour:

- on self and others, with violent temper tantrums if disrupted
- unusual stereotypical movements such as hand flapping and tiptoe gait
- concrete play
- poverty of imagination in play and general activities
- peculiar interests and repetitive adherence
- restriction in behaviour repertoire

Co-morbidities:

- general learning and attention difficulties (about two-thirds)
- seizures (about one-quarter, often not until adolescence).

Autistic spectrum disorder

- **Presents at 2–4 years with impaired social interaction, speech and language disorder and imposition of routines with ritualistic and repetitive behaviour**
- **Usually managed by behaviour modification such as applied behavioural analysis (ABA).**

Slow acquisition of cognitive skills/general learning difficulty

The term 'learning difficulty or disability' (reflecting cognitive learning difficulties) is now preferred to 'mental retardation' or 'mental handicap'. Medical and

educational classification of Intelligence Quotients can be different, with medical models having lower ranges. The educational levels below are useful for general use.

Children with borderline and mild (IQ 70–80) learning difficulties are usually supported by additional helpers (learning support assistants, LSAs) in mainstream schools, whereas children with moderate (IQ 50–70), severe (IQ 35–50) and profound (IQ < 35) learning difficulties are likely to need the resources of special schools.

Severe or profound learning difficulties are usually apparent from infancy as marked global developmental delay, whereas moderate learning difficulties emerge only as delay in speech and language becomes apparent. Mild learning difficulties may only become apparent when the child starts school or much later.

A child with profound learning difficulties will have no significant language and be completely dependent for all of his needs. A child with severe learning difficulties is likely to be able to learn minimal self-care skills and acquire simple speech and language. Both will need high or total supervision and support throughout life.

The prevalence of severe learning difficulty is about 3–4 per 1000 children. Most have an organic cause irrespective of social class, in contrast to moderate learning difficulty (30 per 1000 children) in which children of parents from lower socioeconomic classes are over-represented.

Common causes of developmental delay and learning difficulty are listed in Table 4.3.

Specific learning difficulty

Specific learning difficulty implies the skill described is more delayed than would be expected for the child's level of cognitive ability.

Developmental coordination disorder (DCD) or dyspraxia

Developmental coordination disorder (dyspraxia) is a disorder of motor planning and/or execution with no significant findings on standard neurological examination. It is a disorder of the higher cortical processes and there may be associated problems of perception (how the child interprets what he sees and hears), use of language and putting thoughts together.

The difficulties may impact on educational progress and self-esteem and suggest the child has greater academic difficulties than may be the case. Features include problems with:

- handwriting, which is typically awkward, messy, slow, irregular and poorly spaced
- dressing (buttons, laces, clothes)
- cutting up food
- poorly established laterality
- copying and drawing
- messy eating from difficulty in coordinating biting, chewing and swallowing (oromotor dyspraxia). Dribbling of saliva is common.

Assessment and advice is primarily from an occupational therapist or when necessary a speech and language therapist (oromotor skills/speech). A visual assessment may also be helpful. Dyspraxia in its milder form often goes undetected during the first few years of life as the child achieves gross motor milestones at the normal times. With therapy (emphasis on sensory integration, sequencing and executive planning) and maturity, the condition should improve.

Dyslexia

Dyslexia is a disorder of reading skills disproportionate to the child's IQ. The term is often used when the child's reading age is more than 2 years behind his chronological age. Assessment needs to include vision and hearing and involves an educational psychologist.

Dyscalculia, dysgraphia

These are disorders in the development of calculation or writing skills.

Disorder of executive functions

Executive functions are a collection of cognitive processes that are responsible for activities such as planning and organisation, self-regulation, cognitive flexibility, problem solving and abstract reasoning. They are very necessary to function in and interact with one's social environment. Deficits in executive function can occur as a consequence of acquired brain injury such as those following infection, stroke or trauma. Executive dysfunction may manifest as poor concentration, forgetfulness, volatile mood, overeating and poor social skills.

Associated co-morbidities of specific learning disorders

These are:

- Attention deficit disorder
- Hyperactivity
- Poor sensory integration skills (touch, balance)
- Depression, conduct disorders.

Management of specific learning disorders

Assessment may include vision and hearing and assessment by an occupational therapist, physiotherapist, speech and language therapist and educational psychologist. Co-morbidities need to be identified. Treatment is aimed at improving skill acquisition, with educational and information technology support as appropriate.

Problems with concentration and attention

Attention deficit disorder (ADD) and attention deficit hyperactivity disorder (ADHD) are considered in Chapter 23.

Hearing impairment

Any concern about hearing impairment should be taken seriously. Any child with delayed language or speech, learning difficulties or behavioural problems should have their hearing tested, as a mild hearing loss may be the underlying cause without parents or other carers realising it. Hearing loss may be:

- sensorineural – caused by a lesion in the cochlea or auditory nerve and is usually present at birth
- conductive – from abnormalities of the ear canal or the middle ear, most often from otitis media with effusion.

The causes, natural history and management of hearing loss are listed in Table 4.5. Hearing tests are described in Chapter 3. The typical audiogram in sensorineural and conductive hearing loss is shown in Figure 4.6.

Sensorineural hearing loss

This type of hearing loss is uncommon (1 in 1000 of all live births; 1 in 100 in extremely low birthweight infants). It is usually present at birth or develops in the first few months of life. It is irreversible and can be of any severity, including profound.

The child with severe bilateral sensorineural hearing impairment will need early amplification with hearing aids for optimal speech and language development. Hearing aid use requires close supervision, beginning in the home together with the parents and continuing into school. Children often resist wearing hearing aids because background noise can be amplified unpleasantly. Cochlear implants may be required where hearing aids give insufficient amplification (Fig. 4.7).

Many children with moderate hearing impairment can be educated within the mainstream school system or in partial hearing units attached to mainstream schools. Children with hearing impairment should be placed in the front of the classroom so that they can readily see the teacher. Gesture, visual context and lip movement will also allow children to develop language concepts. Speech may be delayed, but with appropriate therapy can be of good quality. Modified and simplified signing such as Makaton can be helpful for children who are both hearing-impaired and learning-disabled. Specialist teaching and support in preschool and school years is provided by peripatetic teachers for children with hearing impairment. Those with profound hearing impairment may need to attend a school for children who are deaf.

Conductive hearing loss

Conductive hearing loss from middle ear disease is usually mild or moderate but may be severe. It is much more common than sensorineural hearing loss. In association with upper respiratory tract infections, many children have episodes of hearing loss which are usually self-limiting. In some cases of chronic otitis media with effusion, the hearing loss may last many months or years. In most affected children, there are no identifiable risk factors present but children with Down syndrome, cleft palate and atopy are particularly prone to hearing loss from middle ear disease.

Table 4.5 Causes and management of hearing loss

	Sensorineural	Conductive
Causes	Genetic (the majority) Antenatal and perinatal: Congenital infection Preterm Hypoxic-ischaemic encephalopathy Hyperbilirubinaemia Postnatal: Meningitis/encephalitis Head injury Drugs, e.g. aminoglycosides, furosemide (frusemide) Neurodegenerative disorders	Otitis media with effusion (glue ear) Eustachian tube dysfunction: Down syndrome Cleft palate Pierre Robin sequence Mid-facial hypoplasia Wax (only rarely a cause of hearing loss)
Hearing loss	May be profound (>95 dB hearing loss)	Maximum of 60 dB hearing loss
Natural history	Does not improve and may progress	Intermittent or resolves
Management	Amplification or cochlear implant if necessary	Conservative, amplification or surgery

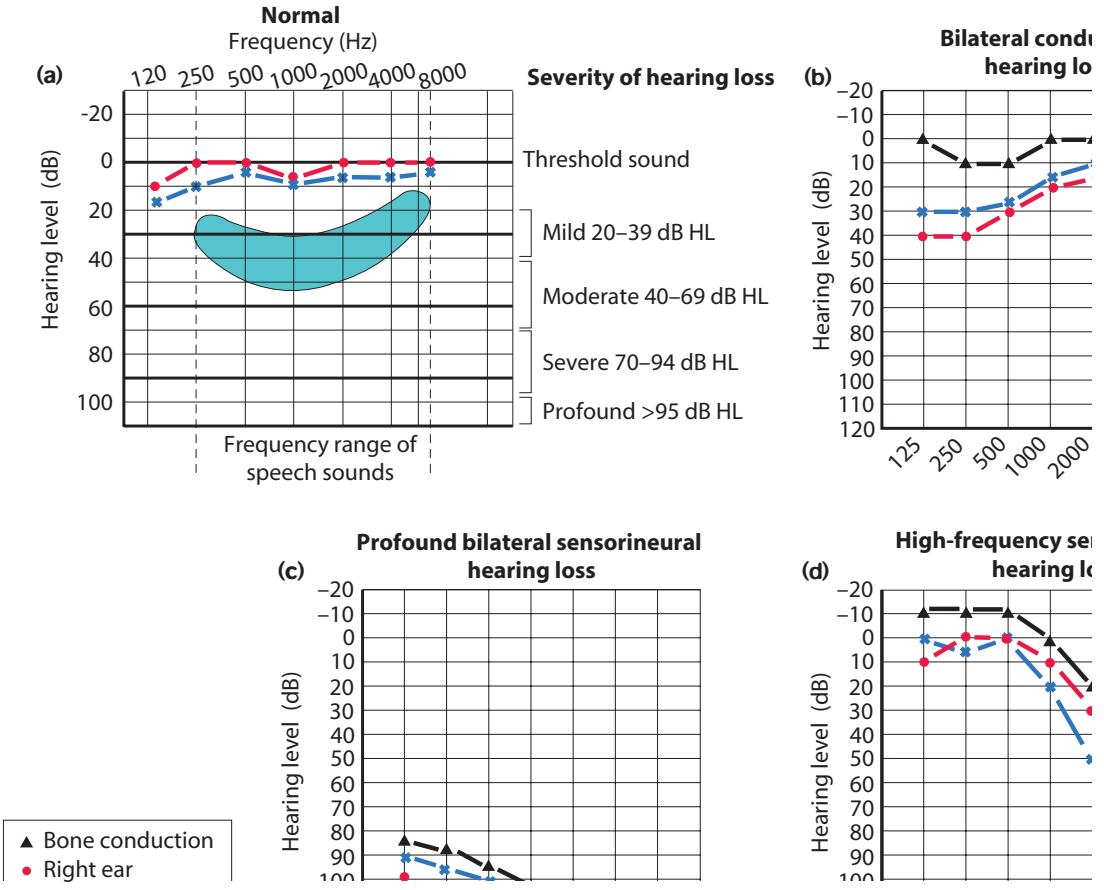


Figure 4.6 **(a)** Audiogram showing normal hearing and the loudness of normal speech (blue area). The consonants are high-frequency sounds, the vowels are low-frequency sounds. **(b)** Audiogram showing bilateral conductive hearing loss. There is a 30–40 dB hearing loss in both the right and left ears. **(c)** Audiogram showing bilateral profound sensorineural hearing loss. **(d)** Audiogram showing bilateral high-frequency sensorineural hearing loss.



Figure 4.7 Cochlear implant. There is a microphone to detect sound, a speech processor and a transmitter and receiver/stimulator. They convert speech into electric impulses which are conveyed to the auditory nerve, bypassing the ear. It provides a deaf person with a representation of sounds.

Summary

Hearing loss

Sensorineural hearing loss:

- is usually present at birth and is irreversible
- for severe hearing impairment, early amplification with hearing aids or cochlear implants is needed for optimal speech and language development
- assistance from peripatetic teachers for children with hearing impairment is required.

Conductive hearing loss:

- is usually due to middle ear disease, often otitis media with effusion
- is usually mild or moderate and transient
- if it does not resolve, insertion of tympanostomy tubes (grommets) with or without the removal of adenoids will need to be considered.

Impedance audiometry tests, which measure the air pressure within the middle ear and the compliance of the tympanic membrane, determine if the middle ear is functioning normally. If the condition does not improve spontaneously, medical treatment (decongestant or a long course of antibiotics or treatment of nasal allergy) can be given. If that fails, surgery is considered, with insertion of tympanostomy tubes (grommets) with or without the removal of adenoids. Hearing aids are used in cases where problems recur after surgery.

The decision whether to intervene surgically should be based on the degree of functional disability rather than on absolute hearing loss.

Any child with poor or delayed speech or language must have their hearing assessed.

Abnormalities of vision

Normal visual development and tests of vision are described in Chapter 3.

Visual impairment may present in infancy with:

- loss of red reflex from a cataract
- a white reflex in the pupil, which may be due to retinoblastoma, cataract or retinopathy of prematurity (ROP).
- not smiling responsively by 6 weeks post-term
- lack of eye contact with parents
- visual inattention
- random eye movements
- nystagmus
- squint
- photophobia.

Squint (strabismus)

In this common condition there is misalignment of the visual axes. The history may be helpful as squints are often intermittent. The parents are usually correct if they report deviation of the eyes. There may be a history of squint in the family. Newborn babies usually have transient misalignments up to 3 months of age. In some infants and young children, marked epicanthic folds may give an appearance of a squint. Any infant with a squint should have red reflexes checked and those persisting beyond 3 months of age should be referred for a specialist ophthalmological opinion. A squint is usually caused by failure to develop binocular vision due to refractive errors, but cataracts, retinoblastoma and other intraocular causes must be excluded.

Squints are commonly divided into:

- *Concomitant* (non-paralytic, common) – usually due to a refractive error in one or both eyes, which is often treated by correction with glasses but may require surgery. These squints are particularly common in children with neurodevelopmental delay. The squinting eye most often turns inwards (convergent), but there can be outward (divergent) or, rarely, vertical deviation.
- *Paralytic* (rare) – varies with gaze direction due to paralysis of the motor nerves. This can be sinister because of the possibility of an underlying space-occupying lesion such as a brain tumour.

Corneal light reflex test

For the non-specialist, the light reflex test is used to detect squints (Fig. 4.8). It is easiest to use a pen-torch held at a distance to produce reflections on both corneas simultaneously. The light reflection should appear in the same position in the two pupils. If it does not, a squint is present. However, a minor squint may be difficult to detect.

Cover test

When a squint is present and the fixing eye is covered, the squinting eye moves to take up fixation (Fig. 4.9). The child's interest can be attracted with a toy or light. The test should be performed with the object near (33 cm) and distant (at least 6 m), as certain squints are present only at one distance. Occlusion

Squints

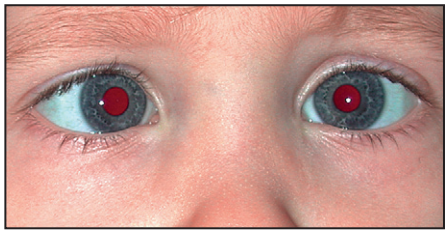


Figure 4.8 Corneal light reflex (reflection) test to detect a squint. The reflection is in a different position in the two eyes because of a small convergent squint of the right eye.

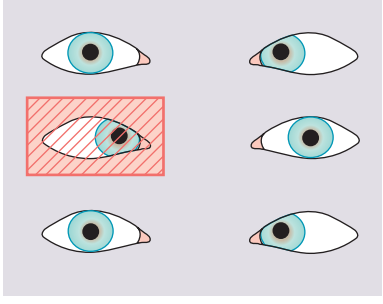


Figure 4.9 The cover test is used to identify a squint. If the fixing eye is covered, the squinting eye moves to take up fixation. This diagram shows a left convergent squint.

Box 4.2 Causes of visual impairment

Genetic	Antenatal and perinatal	Postnatal
Cataract	Congenital infection	Trauma
Albinism	Retinopathy of prematurity	Infection
Retinal dystrophy	Hypoxic-ischaemic encephalopathy	Juvenile idiopathic arthritis
Retinoblastoma	Cerebral abnormality/damage	
	Optic nerve hypoplasia	

should be with a card or plastic occluder. These tests are difficult to perform and reliable results are best obtained by an orthoptist or ophthalmologist.

Refractive errors

Hypermetropia (long sight)

This is the most common refractive error in young children. High degrees or asymmetric hypermetropia should be corrected early to avoid irreversible damage to vision (amblyopia). This is more likely if accompanied by a squint but may occur without.

Myopia (short sight)

This is relatively uncommon in young children and is less likely to cause amblyopia unless it is severe or only one eye is affected.

Astigmatism (abnormal corneal curvature)

High levels of astigmatism may cause amblyopia.

Amblyopia

This is a potentially permanent loss of visual acuity in an eye that has not received a clear image. It affects 2–3% of children. In most cases, it affects one eye; rarely, both are involved. Any interference with visual development may cause amblyopia, such as refractive errors, squint or visual deprivation, e.g. ptosis or cataract. Treatment is by relieving deprivation and correction of any refractive error with glasses, together with patching of the 'good' eye for specific periods of the day to force the 'lazy' eye to work and therefore develop better vision. It is continued until the vision in the 'lazy' eye no longer improves. The longer treatment is delayed, the less likely it is that normal vision will be obtained. Early treatment is essential, as after 7 years of age improvement is unlikely. Considerable encouragement and support often needs to be given to both the child and parents, as young children usually dislike having their eye patched, particularly if vision in the unpatched eye is poor.

Severe visual impairment

This affects 1 in 1000 live births in the UK but is higher in developing countries. A family history of severe visual impairment, developmental delay or extreme prematurity places the infant at an increased risk. The main causes are listed in Box 4.2. In developed countries, about 50% of severe visual impairment is genetic; in developing countries acquired causes such as infection are more prevalent. When visual impairment is of cortical origin, resulting from cerebral damage, examination of the eye, including the pupillary responses, may be normal.

Although few causes of severe visual impairment can be cured, early detection is important, as certain elements may require treatment and much can be done to help the child and parents. Parents of a partially sighted or severely visually impaired child need appropriate advice on how to provide non-visual stimulation using speech and touch, on providing a safe home environment and on how to build the child's confidence. In the UK, advice is usually provided by peripatetic teachers for children with visual impairment. The teachers provide input at both preschool and school ages. Partially sighted children may be able to attend a mainstream school but require special assistance with low vision aids, which include filtered lenses, high-powered magnifiers and small telescopic devices and computers. Severely visually impaired children may need special schooling. Some will need to be taught Braille to enable them to read. While many severely visually impaired children have a visual disability alone, at least half have additional neurodevelopmental problems.

Summary

Regarding vision

- Abnormal eye movements in a newborn infant or not smiling responsively by 6 weeks post-term – can the infant see?
- Any infant with a squint (fixed or otherwise) after 3 months of age – refer for an ophthalmological opinion.
- Concomitant squint – common, usually due to a refractive error in one or both eyes.
- Paralytic squint – rare, due to paralysis of the motor nerves; if rapid onset, consider space-occupying lesion.
- Testing for squints – corneal light reflex (reflection) test for the non-specialist, cover test for the specialist.

Multidisciplinary child development services

Although children with a wide range of conditions have additional needs, the term 'special needs' is usually used for children with developmental problems and disabilities. In order to optimise their assessment and care on an ongoing basis, child development services have been developed nationally on a geographic area as a secondary care service.

A child development service (CDS):

- is multidisciplinary with predominantly health professionals (paediatrician, physiotherapist, occupational therapist, speech and language therapist, clinical psychologist, specialist health visitor, dietician) in the team but often also includes a social worker (Fig. 4.10)
- is multi-agency (Fig. 4.11) and may include health, social services, education, volunteers, voluntary agencies, parent support groups
- aims to provide a coordinated service with good inter-agency liaison to meet the functional needs of the child
- predominantly sees preschool children with moderate or severe difficulties but may have resources to support children with milder problems
- may provide multidisciplinary support and monitor children up to school-leaving age (16–19 years)
- maintains a register of children with disabilities and special needs (this may be held by Social

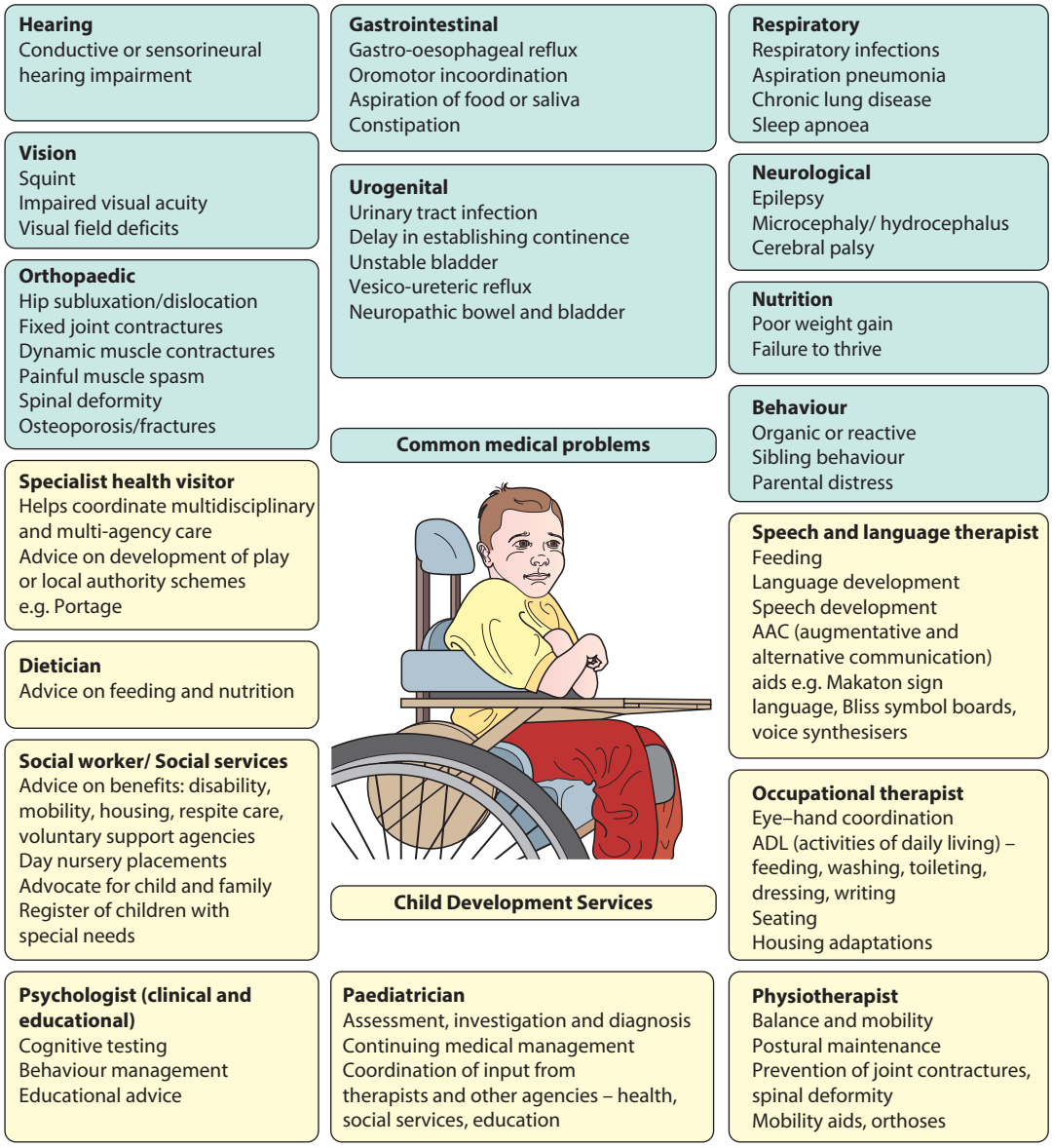


Figure 4.10 Common medical conditions and the many professionals in the child development services involved in the care of children with developmental problems.

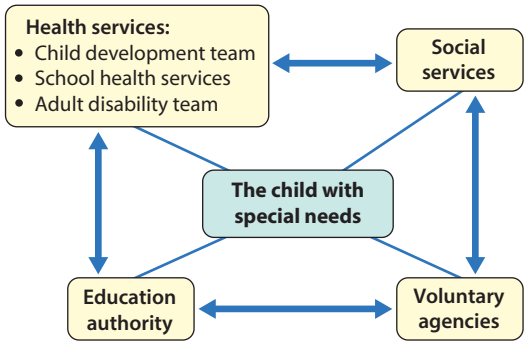


Figure 4.11 Children with special needs are supported by the integrated input of health and social services, local education authorities and voluntary agencies.



Figure 4.12 An example of a touch screen speaking communication aid to assist children who may have speaking and movement difficulties.

Services, but there is an increasing trend to single multi-agency Special Needs registers)
- is community or hospital based but has emphasis on children's needs within the community (home, nursery, school), regardless of its location
- often has a nominated key worker for a child, to facilitate parents getting access to information and services their child may need.

Child development services in the UK now usually use the Common Assessment Framework (CAF) to allow multidisciplinary sharing of information.

Emphasis is on:

- diagnosis
- assessment of functional skills
- provision of therapy
- regular review
- a coordinated approach to care (multidisciplinary, multi-agency).

Functional skills kept under review include:

- mobility
- hand function
- vision
- hearing
- speech, language and communication, including social/communication skills
- behaviour, social and emotional skills
- self-help skills, including continence
- learning.

Many children with special needs have medical problems (Fig. 4.10) which require investigation, treatment and review. Good inter-professional communication is vital for well-coordinated care. This is assisted by all professionals keeping entries in the child's personal child health record up to date.

In addition to locally organised child development services, specialist neurodisability services are required for:

- Rehabilitation following acquired brain injury
- Surgery for cerebral palsy, scoliosis
- Gait analysis
- Spasticity management, including botulinum toxin injections to muscles
- Epilepsy unresponsive to two or more anticonvulsants or where there is severe cognitive and behavioural regression related to epilepsy
- Complex communication disorders, diagnosis and therapeutic intervention
- Mixed complex learning problems, often with neuropsychiatric co-morbid symptoms
- Provision of communication aids (Fig. 4.12).
- Sensory impairments, e.g. cochlear implants
- Services for severe visual and hearing impairment
- Specialised seating/wheelchairs and orthoses (Fig. 4.13).
- Management of movement disorders, e.g. continuous infusion of intrathecal baclofen and deep brain stimulation to basal ganglia.

Needs are likely to change over time with key stages being at transition to school and adult services. A care plan should be developed at each stage and needs to be shared with the child and family and then regularly reviewed. Involvement with specialist services may be of variable frequency throughout childhood. Collaboration across services is vital in promoting a service tailored around the child and family.

Summary

Children with developmental problems and disabilities

- are looked after by local multidisciplinary child development services
- often have complex medical needs
- need regular review, as needs change with time affecting a child's activity and participation
- require coordination of care between the family and the many professionals involved, as well as close liaison with education and social services.



(a)



(b)

Figure 4.13 (a) A boy with spastic cerebral palsy is able to walk with the help of a frame. **(b)** Motorised wheelchair which enables this young person with cerebral palsy to be mobile.

Education

In England and Wales, several Education Acts and the 2000 Code of Practice have made provision for children with special educational needs to receive educational input appropriate to their requirements. This includes their right to integration into mainstream education whenever possible. Education authorities have a duty to identify children whose special educational needs will require additional resources.

Initial recognition that a child may have special educational needs (SEN) may occur at the preschool stage in children with specific or global developmental delay or with specific disabilities, or may only become evident when the child is of school age. Early identification and initiating appropriate help maximises the child's opportunity to progress. The local education authority (LEA) is informed of these children by a process called 'notification', usually by the paediatrician if preschool or by an education professional if at school. Notification is a statutory requirement if a health professional identifies that a child may have special educational needs.

The 2000 Code of Practice suggests that pupils' needs may fall within four broad areas:

- Cognition and learning
- Communication and interaction
- Behaviour, emotional and social development
- Sensory and/or physical needs.

Where a child is identified as having special educational needs, a special educational needs' coordinator (SENCO) within the teaching staff is responsible for formulating an individual education plan (IEP) for the child within the school or nursery or for seeking help from external services, e.g. educational psychology.

Extra educational provision in school is commonly provided at three levels: *School action* is the first internal level of extra support; *School action plus* may involve external support services, such as a behaviour specialist or teacher for the visually impaired. If a child's needs are more significant, the LEA is asked to conduct a statutory assessment of a child's educational needs. This may lead to a Statement of Special Educational Needs, which is a document identifying the extra help the child needs and is reviewed annually. Parents can also request such an assessment. The statementing process may start before school age in preparation for a child who is expected to have SEN, such as children with Down syndrome or severe cerebral palsy.

Although children are integrated into mainstream school where practicable, special schools or units are usually more suitable for children with severe learning difficulties and sometimes for those with severe physical, sensory, communication or behaviour problems. Many special educational placements will have a need for therapy input (physiotherapy, occupational and speech/language therapy) as well as specialised teaching resources. Support for the behavioural needs of a child may come from a clinical or educational psychologist.

Transition of care to adult services

In the UK, adult disability services are in general still poorly developed by comparison to those provided for children. Young adults with severe learning and physical disabilities are supported by Adult Learning Disability Teams, but there is only limited national provision for those with mild or moderate learning disabilities or with a predominantly physical disability. Major issues for young adults with disabilities include social challenges: namely, care, housing, mobility, finance, leisure, employment, and genetic and sexual counselling. Health information must be properly transferred from child to adult health services if reinvestigation of already well-clarified conditions is to be avoided.

The rights of disabled children

Irrespective of their disability, the aspirations and rights of children, as affirmed by the United Nations Convention on the Rights of the Child, need to be respected (see Ch. 1). Technological advances to improve mobility, communication and emotional expression are helping enable people with disability to better achieve their full potential, rather than being held back by their disability. However, this requires skilled assistance and adequate resources. Prominent public figures who function effectively despite disabilities help to make the public appreciate what can be achieved and serve as an inspiration to those with disabilities. The World Health Organization's model of the international classification of functioning, disability and health (ICF) stresses the important outcomes of activity and participation. Any interventions for people with disability either on an individual level or in society as a whole should aim to improve these outcomes.

Further reading

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5

Care of the sick child

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Most sick children are cared for by their parents at home. Medical management is initially given by general practitioners or, in some countries, primary care paediatricians. Most hospital admissions are at secondary care level. A smaller number of children will require tertiary care in a specialist centre, e.g. paediatric intensive care unit, cardiac or oncology unit. Most specialist centres now share care within clinical networks, with the centre linked to a number of surrounding hospitals. For very rare and complex treatments, e.g. organ transplantation and craniofacial surgery, there are a few national centres (Fig. 5.1).

Primary care

The majority of acute illness in children is mild and transient (e.g. upper respiratory tract infection, gastroenteritis) or readily treatable (e.g. urinary tract infection). Although serious conditions are uncommon (Fig. 5.2), they must be identified promptly. The condition of sick children, especially infants, may deteriorate rapidly, and parents require rapid access to a general practitioner or other healthcare professionals working in primary care, who in turn require ready access to secondary care. Advice may also be obtained from a health professional by telephone, e.g. via NHS Direct, via the internet with NHS Direct Online or by NHS Direct cable TV services. Although an individual general practitioner will care for relatively few children with serious chronic illnesses (e.g. cystic fibrosis, diabetes mellitus) or disability (e.g. cerebral palsy), each affected child and family are likely to require considerable input from the whole of the primary care team.

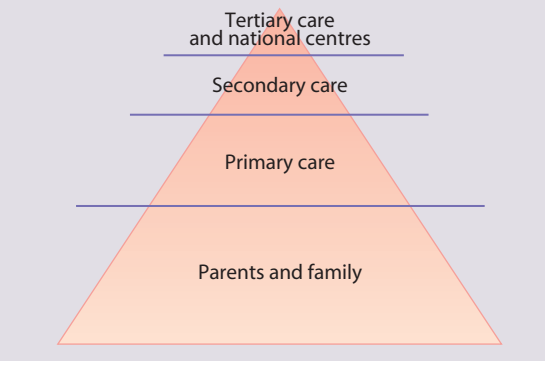


Figure 5.1 Schematic representation of the 'clinical iceberg' of the provision of care for sick children. (Adapted from Audit Commission, Children First, 1993.)

Hospital care

Accident and Emergency

Approximately 3 million children (1 in 4) attend an Accident and Emergency (A&E) department each year in England and Wales. The services which should be provided for children are shown in Box 5.1. The number

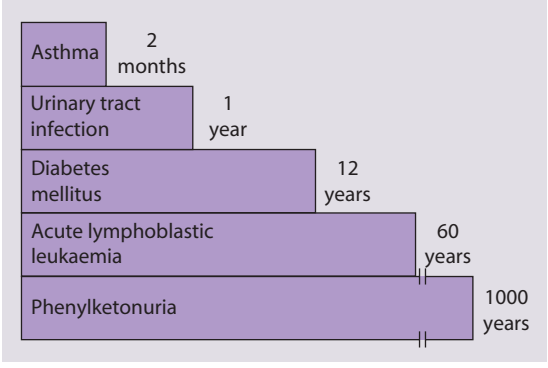


Figure 5.2 Number of years a general practitioner needs to work before encountering a child newly presenting with these conditions.

Box 5.1 Services which should be available for children attending an Accident and Emergency (A&E) department

Environment	Staff	Medical care
Separate waiting area, play facilities, child friendly treatment and recovery areas	Medical and nursing staff trained and experienced in the care and treatment of children	Resuscitation and other equipment for children. Priority for prompt treatment
Access for parents to examination, X-ray and anaesthetic rooms	Non-paediatric staff trained in communicating with children and families	Rapid transfer if inpatient admission is needed.
	Effective communication with other health professionals	Child protection policies in place
		Procedures and counselling in place following the sudden death of a child

Adapted from *Welfare of Children in Hospital*, HMSO, London, 1991.

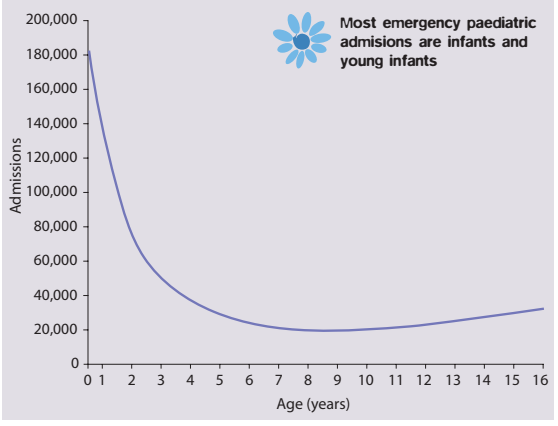


Figure 5.3 Hospital inpatient admission for children and young people aged 0–16 years in England. (Adapted from *Trends in Children and Young People's Care: Emergency Admission Statistics, 1996/97–2006/07, England.* Department of Health, London.)

of departments able to meet these expectations is increasing, often by creating a dedicated children's A&E department.

Hospital admission

In England and Wales, 1.9 million, i.e. 1 in 6 children is admitted to hospital each year, representing 16% of all hospital admissions. About 60% of acute admissions are under the care of paediatricians, and the remainder are surgical patients (although a paediatrician is also involved in their care while they are in hospital, to oversee any medical requirements) (Fig. 5.3). Most paediatric admissions are of infants and young children under 5 years of age and are emergencies, whereas surgical admissions peak at 5 years of age, one-third of which are elective. The reasons for medical admission are shown in Table 5.1.

Although primary and community health services for children have improved markedly over the last decade, the hospital admission rate has continued to rise, although it may now be plateauing (Fig. 5.4). The reasons for this are unclear, but probably include:

- Lower threshold for admission: there appears to be an increased expectation of hospital admission by parents and medical staff worried that the child's clinical condition may deteriorate
- Repeated hospital admission of children with complex conditions who would have died in the past but are now surviving, e.g. very low birthweight infants from neonatal intensive care units, children with cancer or organ failure.

Table 5.1 Reason for emergency admission of children <15 years old to hospital

System	Specific disorders
Respiratory 25%	Respiratory infections 20%
	Asthma 3%
Injuries and poisoning 17%	Head injury 5%
	Poisoning 1%
Gastroenterological 13%	Gastroenteritis 5%
Infection 6%	Viral infection 5%
Urogenital 3%	Urinary tract infection 2%
Neurological 2%	Seizures 1%
Endocrine and metabolic 2%	Diabetes mellitus 1%
Skin 2%	
Muskuloskeletal 2%	
Other 28%	

Data based on 58 061 admissions, 2009–2010. ISD, Scotland.

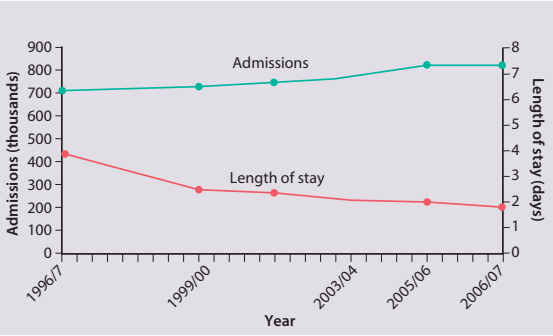


Figure 5.4 There has been an increase in the number of children and young people (0–19 years) admitted to hospital. This is because of a marked increase in the number of paediatric admissions, whereas surgical admissions have declined slightly. However, the average length of stay continues to fall. (Adapted from *Trends in Children and Young People's Care: Emergency Admission Statistics, 1996/97–2006/07, England.* Department of Health, London.)



Figure 5.5 Providing terminal care in a child's home. Although this child required a subcutaneous morphine infusion to control her pain from malignant disease, she was able to remain at home and enjoyed playing with her pet rabbit. (By kind permission of her parents and Dr Ann Goldman.)

Strenuous efforts are made to reduce the rate and length of hospitalisation (Fig. 5.4):

- Speciality of ambulatory paediatrics has been developed; it encompasses specialist paediatricians providing hospital care for immediate medical problems outside inpatient paediatric wards
- Dedicated children's short stay beds within or alongside the A&E department are increasingly available to allow children to be treated or observed for a number of hours and discharged home directly, avoiding the need for admission to the ward
- Day-case surgery has been instituted for many operations which used to require overnight stay.
- Day units are used for complex investigations and procedures instead of inpatient wards
- Shared care may be provided between hospitals and primary care, with paediatricians and other healthcare professionals seeing children at home or in primary care settings
- Homecare teams aim to provide care in the child's home and thereby reduce hospital attendance, admission and length of stay. Most teams comprise community paediatric nurses, but some include doctors, and either cover all aspects of paediatric care within a geographical area or are for a specific condition, e.g. cystic fibrosis or malignancy, usually centred around a tertiary referral centre. The problems managed at home by such teams include:
  - Changing postoperative wound dressings or managing burns
  - Day-to-day management and support for the family for chronic illnesses, e.g. diabetes mellitus, asthma and eczema
  - Specialist care, e.g. home oxygen therapy, intravenous infusions via a central venous catheter (e.g. antibiotics or chemotherapy) or peritoneal dialysis
  - Symptom and pain control and emotional support of terminally ill children (Fig. 5.5)
- Children's hospices provide respite or terminal care for children with life-threatening conditions, including malignancy disease, neurodegenerative, metabolic and other disorders.
- Some teams provide a 'hospital at home' service for children who are acutely ill, in order to avoid hospitalisation.

Hospital admission of children:

- **Hospital admission of children should be avoided whenever possible.**
- **Most medical admissions are infants and young children; surgical admissions occur throughout childhood.**

Children in hospital

Children should only be admitted to hospital if their care cannot be provided safely at home. Removing young children from their familiar environment to a strange ward is stressful and frightening for the child, parents and family. Ill or injured children may regress in their behaviour, acting younger than their actual age. It also disrupts family routines, not only of the child in hospital but also of siblings who still need to be looked after at home and transported to and from nursery or school.

Family-centred care

Care in hospital should be child- and family-centred. Parents and siblings should be involved in the child's care, which should be appropriate for the child's

physical and emotional maturity and needs. A holistic approach should be adopted towards the child and his family rather than simply focusing on the medical condition. Young children may interpret the pain experienced in hospital and separation from their home or parents as punishment. In general, the distress arising from separating children from their mothers is greatest in young children, and increases the longer the length of stay and the more frequently the child is admitted. Parents of infants and young children should be encouraged to stay with their child overnight and continue to provide the care and support they would give at home. Parents know best about their child's usual behaviour and habits and due attention must be paid to their worries or comments. Many parents rapidly learn some of the nursing skills, e.g. tube feeding, required by their child. Good communication is needed between staff and parents to arrive at a mutually agreed plan of responsibilities for looking after the child. This will avoid parents either feeling pressurised to accept responsibilities they are not confident about or feeling brushed aside and undervalued by staff. Parents should be able to stay overnight with their child.

Child-orientated environment

Children should be cared for within a children's ward. Adolescents should be with others of their own age and not forced to accept ward arrangements designed for babies or adults. Education and facilities for play should be provided.

Information and psychosocial support

Detailed information should be provided, given personally and preferably also written and available in appropriate ethnic languages. Staff should be sensitive to the family's individual needs according to their social, educational, cultural and religious background. Play specialists should be part of the ward team because they can help children understand their illness and its treatment through play. Emotional and psychological support should be available. For elective admissions, children and their families should be offered an advance visit and have details of proposed treatment and management explained at an appropriate level.

Skilled staff

Children in hospital should be cared for by specially trained medical, nursing and support staff. Every child admitted to hospital should be supervised by a children's physician or surgeon. Children constitute only a relatively small proportion of the workload in acute surgical specialities, so surgeons and anaesthetists should treat a sufficient number of children to maintain their skills. There should be a 'named nurse' responsible for planning and coordinating care by other nurses to ensure that families receive all the information they need and provide a link with staff involved in discharge planning and post-discharge arrangements.

Multidisciplinary care

Successful management of paediatric conditions often relies on a network of multidisciplinary care, with all the professionals working well together as a coordinated team. If this breaks down, particularly when dealing with complex issues such as child protection and long-term disability, the consequences may be disastrous for the child, family and professionals involved. Child psychiatrists, the community paediatric team and social services are important members of the team.

Tertiary care

As the number of children requiring tertiary care is relatively small, it is concentrated in specialist centres. Increasingly, the centre is linked to several district general hospitals to form a clinical network. These centres have the advantage of having a wide range of specialists, not only medical staff but also nursing and other healthcare professionals, and diagnostic and other services. A disadvantage is that they are often some distance from the child's home and hospital stay may be prolonged, e.g. following a bone marrow transplant. Accommodation for parents should be provided. Shared care arrangements between tertiary centres and local hospitals are designed to minimise the need for the child to travel to the specialist centre, but depend on excellent communication to be effective and valued by the family. For example, a child with leukaemia would attend a tertiary centre for the initial diagnostic assessment and treatment, and subsequently for specialised treatment and periodic review, but much of the maintenance therapy would be provided by the local hospital together with monitoring of their health and regular blood and other tests performed by a specialist nurse at home.

Summary

Children in hospital should be provided with:

- Family-centred care: holistic approach to family, parent able to stay and provide parental care
- Child-oriented environment: appropriate for child's age, together with education and play facilities
- Information and psychosocial support: verbal and written information for both parents and child
- The opportunity for children and families to express their views and fears and be listened to
- Skilled staff: specially trained to care for children
- Multidisciplinary care
- Access to tertiary care: with shared care arrangements with local hospital and primary care.

Pain

It is easy to ignore or underestimate pain in children. Pain should ideally be anticipated and prevented.

Acute pain

This may be caused by:

- Tissue damage, e.g. burns or trauma
- Specific disease process, e.g. sickle cell crisis
- Medical intervention, i.e. investigations or procedures
- Surgery.

Chronic pain

In children, chronic severe pain sometimes occurs as a result of disease such as malignant disease or juvenile idiopathic arthritis. Intermittent pain of mild or moderate severity, e.g. headache or recurrent abdominal pain, is more common.

Older children can describe the nature and severity of the pain they are experiencing. In younger children, assessing pain is more difficult. Observation and parental impression are commonly used and a number of self-assessment tools have been designed for children over 3 years old (Fig. 5.6).

Management

The approaches to pain management are listed in Box 5.2. This should allow pain to be prevented or kept to a minimum. Age-appropriate explanation should be given when possible and the approach be reassuring; however, it is imperative not to lie to children, otherwise they will lose trust in what they are told in the future. Distraction techniques such as blowing bubbles, telling stories, holding family toys or playing computer games, as well as the involvement of trained play specialists, can be highly successful in ameliorating pain in children. Some children develop particular preferences for a particular venepuncture site or distraction technique, and this should be accommodated as far as possible.

For minor medical procedures, e.g. venepuncture or inserting an intravenous cannula, pain can be alleviated by explanation and the use of a topical anaesthetic. Additional and appropriate use of inhalation agents such as nitrous oxide (laughing gas) or the adjunctive use of mild sedation alongside pain relief, e.g. intranasal midazolam, can be helpful for more painful procedures such as suturing a wound. For more invasive procedures, e.g. bronchoscopy, a general anaesthetic should be given.

Postoperative pain can be markedly reduced by local infiltration of the wound, nerve blocks and postoperative analgesics. For severe pain, there was reluctance in the past to use morphine in children for fear of depressing breathing. This should not occur when morphine is given in appropriate dosage under nursing supervision to children with a normal respiratory drive. Intravenous morphine can be given using a patient-controlled delivery system in older children or a nurse-controlled system in young children.

Pain should be anticipated and prevented rather than treated.

Box 5.2 Approaches to pain management

Explanation and information

- Psychological, by the parent, doctor, nurse or play specialist
- Behavioural
- Distraction
- Hypnosis

Medical

- Local: anaesthetic cream, local anaesthetic infiltration, nerve blocks, warmth or cold, physiotherapy, transcutaneous electrical nerve stimulation (TENS)
- Analgesics:
  - Mild – paracetamol, NSAIDs
  - Moderate – codeine, NSAIDs
  - Strong – morphine
- Sedatives and anaesthetic agents:
  - Intranasal midazolam, nitrous oxide, general anaesthetic
- Anti-epileptic and antidepressant drugs for neuropathic pain

Consider the route for analgesics – oral if possible, otherwise intravenous, subcutaneous or rectal.
(NSAIDs, non-steroidal anti-inflammatory drugs).

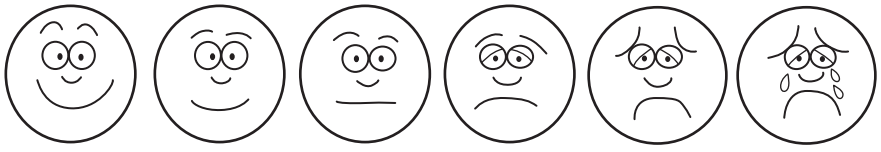


Figure 5.6 An example of a scoring system for pain assessment in children. Wong Baker Faces scale. (From Wong DL et al. 2001. *Wong's Essentials of Pediatric Nursing,* St Louis, MO. Copyright Mosby. Reprinted with permission).

Prescribing medicines for children

There are marked differences in the absorption, distribution and elimination of drugs between children and adults.

Absorption

In the neonate and infant, oral formulations of drugs are given as liquids. However, their intake cannot be guaranteed and absorption is unpredictable as it is affected by gastric emptying and acidity, gut motility and the effects of milk in the stomach. In acutely ill neonates and infants, drugs are given intravenously to ensure reliable and adequate blood and tissue concentrations. Intramuscular injections should be avoided if at all possible as there is little muscle bulk available for injection, absorption is variable and they are painful. Rectal administration can be used for some drugs; absorption is more reliable, but this route is not popular in the UK. Significant systemic absorption can occur across the skin, particularly in pre-term infants. Occasionally this can be used therapeutically, but is a potential cause of toxicity, e.g. alcohol and iodine absorption from cleansing solutions applied to the skin for procedures.

Young children find it difficult to take tablets and a liquid formulation is required. Most are glucose-free. Persuading children to take medicines is often a problem, especially if the preparation has an unpleasant taste; experience and imagination help to overcome their reluctance. Adherence (compliance) is improved when medicines are only required once or twice a day and if regimens are kept simple.

Distribution

Water comprises a larger percentage of the body in the neonate (80%) than in older children and adults (55%). Drugs which distribute within the extracellular fluid will require a larger dose relative to body weight in infants than in adults. As extracellular fluid correlates with body surface area, this is used when accurate drug dosage is required, e.g. cytotoxic agents. For drugs with a high margin of safety, drug dosages are expressed per kilogram body weight or based on age, with the assumption that the child is of average size. Weight-based dosages should not simply be extrapolated to older children, as the dosage will be excessively large.

In the first few months of life, the plasma protein is low. More of a drug may be unbound and pharmacologically active. In jaundiced babies, bilirubin may compete with some drugs, e.g. sulphonamides, for albumin binding sites, making such drugs unsuitable for use in this situation.

Elimination

In neonates, drug biotransformation is reduced, as microsomal enzymes in the liver are immature. This leads to a prolonged half-life of drugs metabolised in the liver, e.g. theophylline. Renal excretion is reduced by the low glomerular filtration rate, which increases the half-life of some drugs, e.g. vancomycin. Measuring the plasma drug concentration is necessary under these circumstances.

Summary

Regarding medicines for children

- Oral formulations need to be given as liquids in infants and young children
- Medicines are usually prescribed per kilogram of body weight, but check the maximum dose
- Intramuscular drugs should be avoided if at all possible
- Intravenous drug dosages can easily be miscalculated as they vary widely in children because of their different size, and drugs often need to be diluted; all dosages and dilutions must be checked independently by two trained members of staff
- To improve compliance, use formulations requiring the least number of times to be taken per day
- Always check drug dosage in the BNF (*British National Formulary*) for children.

Breaking bad news

Doctors often face the difficult task of imparting bad news to parents and children. In paediatric practice, it may be because there is:

- a serious congenital abnormality at birth, e.g. chromosomal disorder
- the diagnosis of a disabling condition, e.g. cerebral palsy, neurodegenerative disorder, gross intracranial abnormality seen at ultrasound in pre-term infants
- a serious illness, e.g. meningitis or malignant disease, or an accident, e.g. head injury
- sudden death of a child, e.g. sudden infant death syndrome (SIDS).

Initial interview

The manner in which the initial interview is conducted is very important. It may have a profound influence on the parents' ability to cope with the problem and their subsequent relationship with health professionals. Parents often continue to recall and recount, for many years, details of the initial interview when they were informed that their child had a serious problem. Parents of children with life-threatening illnesses have said that what they valued most was open, sympathetic, direct and uninterrupted discussion in private that allowed sufficient time for doctors to repeat and clarify information and for them to ask questions (Box 5.3).

Discharge from hospital

Children should be discharged from hospital as soon as clinically and socially appropriate. Although there is increasing pressure to reduce the length of hospital stay to a minimum, this must not allow discharge planning to be neglected. Before discharge from hospital, parents and children should be informed of:

- The reason for admission and any implications for the future

Box 5.3 How parents wish to be told the diagnosis of a life-threatening illness

Setting
- In private
- Uninterrupted
- Unhurried
- Both parents (or friend/relative) present if possible
- Senior doctor
- Nurse or social worker present

Establish contact
- Find out what the family knows or suspects
- Respect the family's vulnerability
- Use the child's name
- Do not avoid looking at them
- Be direct, open, sympathetic

Provide information
- Flexibility is essential
- Do not protect from bad news, but pace giving it
- Name the illness
- Describe symptoms relevant to child's condition
- Discuss aetiology – parents will usually want to know
- Anticipate and answer questions. Do not avoid difficult issues because parents have not thought to ask

Explain long-term prognosis
- If the child is likely to die, listen to concerns about time, place and nature of death
- Outline the support/treatment available

Address feelings
- Be prepared to tolerate reactions of shock, especially anger or weeping
- Acknowledge uncertainty
- How is it likely to affect the family?
- What and how to tell other children, relatives and friends?

Concluding the interview
- Elicit what parents have understood
- Clarify and repeat
- Acknowledge that it may be difficult for parents to absorb all the information
- Mention sources of support
- If possible, give parents contact telephone number
- Give address of self-help group

Follow-up
- Offer early follow-up
- Suggest to families that they write down questions in preparation for the next appointment
- Ensure adequate communication of content of interview to:
  - other members of staff
  - general practitioner and health visitor
  - other professionals, e.g. a referring paediatrician.

Adapted from Woolley H, Stein A, Forrest GC et al. 1989. Imparting the diagnosis of life-threatening illness in children. *British Medical Journal* 298:1623–1626.

- Details of medication and other treatment
- Any clinical features which should prompt them to seek medical advice, and how this should be obtained
- The existence of any voluntary self-help groups if appropriate
- Problems or questions likely to be asked by other family members or in the community. These should be anticipated by the doctor and discussed. What do the nursery or school, baby-sitters or friends need to know? What about sports, etc?

In addition:

- Suitability of home circumstances needs to be assessed, particularly when the home requires adaptation for special needs
- Social support may need to be arranged, especially in relation to child protection
- Medical information should be added to the child's personal child health record
- Consider which professionals should be informed about the admission and what information it is relevant for them to receive. This must be done before or at the time of discharge. The aim is to provide a seamless service of care, treatment and support, with the family and all the professionals fully informed (Fig. 5.7). This can be facilitated for children with a chronic illness or disability by having a key worker to coordinate their care.

Ethics

Situations arise in paediatric practice in which the course of action that should be followed is unclear. Knowledge of the ethical theories and principles which underpin medical practice is helpful in understanding the issues involved. It is important to justify decisions to investigate or treat in accordance with these principles, and in language that is clear to all concerned.

Definitions of the principles of medical ethics

- *Non-maleficence* – do no harm (psychological and/or physical)
- *Beneficence* – positive obligation to do good (these two principles have been part of medical ethics since the Hippocratic Oath)

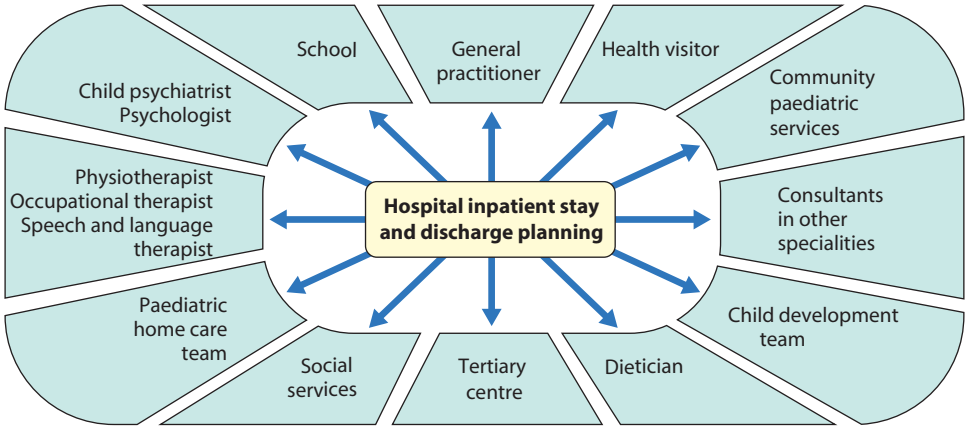


Figure 5.7 Some of the professionals who may need to be informed on admission or discharge about a child admitted to hospital.

- *Justice* – fairness for all, equity and equality of care
- *Respect for autonomy* – respect for individuals' rights to make informed and thought-out decisions for themselves in accordance with their capabilities
- *Truth-telling and confidentiality* – important aspects of autonomy that support trust, essential in the doctor–patient relationship
- *Duty* – the moral obligation to act irrespective of the consequences in accordance with moral laws which are universal, apply equally to all and which respect persons as autonomous beings
- *Utility* – the obligation to do the greatest good for the greatest number
- *Rights* – justifiable moral claims, e.g. the right to life, respect, education, which impose moral obligations upon others.

Application of ethical principles to paediatrics

Non-maleficence

Children are more vulnerable to harm. This includes their suffering from fear of procedures, which they may be too young to express verbally. Doctors may do harm from lack of skill or knowledge, especially if they do not treat children frequently.

Beneficence

The child's interest is paramount. In the UK, this is enshrined in the Children Act 1989 and the UN Convention on the Rights of the Child. This may sometimes conflict with parental autonomy, such as the emergency treatment of a child where the parent is not immediately available or when details are given to social workers in suspected child abuse.

Justice

This involves ensuring a comprehensive child health service, including the prevention of illness, and equal access to healthcare, even when poverty, language barriers and parental disability are present.

Autonomy

Children have restricted but developing rights in law. Parents are trusted to make decisions on their child's behalf because they will usually act in the child's best interests, but there may be circumstances, e.g. child abuse, in which this is not the case.

Truth-telling

It is more difficult with children than adults to be sure that they understand what is happening to them. For example, it is easy to reassure children falsely that procedures will not hurt; when they find this is untrue, trust will be lost for future occasions.

Consent

Valid consent is required for all medical interventions other than emergencies or when urgent intervention is necessary to prevent serious risk of present or future harm. It provides the ethical and legal authority for action which would otherwise be a common assault or interfere with the right of individuals to decide what should be done to them (autonomous choice). To be valid, consent must be sufficiently informed, and freely given by a person who is competent to do so. Clinicians have a duty to provide sufficient information to enable a reasonable person to make the decision and must answer all questions honestly. Information has to be given in language that is clear and understandable. In UK law, the legal age of consent to medical treatment is 16 years. The right of children below this age to give consent depends on their competence rather than their age. They may consent to medical examination and treatment provided they can demonstrate that they have the maturity and judgement to understand and appraise the nature and implications of the proposed treatment, including the risks and alternative courses of action. This is known as Gillick competence.

When a child lacks the maturity and judgement to give consent, this capacity is given to a person having parental responsibility – usually a natural parent, or to

a court. In practice, problems occur only when there is disagreement between the child and the parents and clinicians over treatment, e.g. contraception for under 16-year-olds.

When a girl less than 16 years of age requests contraception without parental knowledge, a professional can provide it if satisfied that she cannot be persuaded to inform her parents, that she is likely to have sex with or without contraception and that receiving contraception is in her best interests. These are known as the Fraser guidelines.

Despite including children's views in consent, legal judgements have not supported children who refuse treatment parents and clinicians feel to be in their best interests, especially if its purpose is to save life or prevent serious harm, e.g. heart transplantation for acute cardiomyopathy in an intelligent 15-year-old patient. Where disputes cannot be resolved by negotiation or mediation, or there is doubt over the legality of what is proposed, legal advice should be sought. Whatever the outcome, children should have their views heard and be given reasons as to why they are being over-ridden.

Confidentiality

Children are owed the same duty of confidentiality as adults, irrespective of their legal capacity. In general, personal information about them should not be shared without their consent or agreement unless it is necessary for their health or to protect them from serious harm, e.g. in actual or suspected child abuse.

Best interests

It is a general ethical and legal maxim that the best interests of the child are paramount. Doctors therefore have a duty to save life, restore health and prevent disease by treatments that confer maximum benefit and minimal harm and which respect the autonomy of the child as far as possible. Parents have the ethical and legal duty to make decisions on behalf of their child, provided that they act in their best interests. Disputes may arise over what constitutes best interests and who should decide about them; they may require legal intervention, especially when the withholding or withdrawing of life-sustaining treatment is involved. Courts have generally been supportive of the position that in some circumstances the burden to the child of providing life-sustaining treatment outweighs its benefits.

Case Histories 5.1 and 5.2 demonstrate some of the ethical problems encountered in paediatrics.

The ethics of research in paediatrics

Research involving children is important in promoting children's health and well-being and may provide an evidence base for practice. Children differ from adults in their anatomy, physiology, disease patterns and responses to therapy but many drugs in current use have not been tested on them. However, children are perhaps more vulnerable to the harm which may be produced by research and should be protected against it.

Case History

5.1 Meningococcal septicaemia

Jack, aged 5 years, has a fever and purpuric rash and you suspect he may have potentially fatal meningococcal septicaemia. Jack hates needles and makes it clear that he rejects any sort of injection. 'No I don't want an injection, go away' is the message, loud and clear, when you try to take blood, do a lumbar puncture, and insert an indwelling intravenous cannula for his antibiotics. Yet with the full and anxious approval of his parents, you go ahead and do these things anyway. But if Jack was 25 years old and made it clear that he refused your interventions, while you'd strongly urge him to give permission and explain that he was in real danger of dying as a result of such refusal, you would not (presumably) treat him against his will, even if his mother and father still urged you to do so.

In contrast to normal adult medical ethics, in paediatrics the autonomy of the patient either is not present at all (as in babies and young infants) or is often not sufficiently developed to be respected if the child's decision conflicts with what appropriate other people consider to be in that child's best interests. The decisions about the child's medical care are generally entrusted to the parents. Why the parents? They are given the privilege and responsibility of making decisions on behalf of their children largely because they are most likely to protect and promote the interests of their children. The normal assumption in paediatric practice is that doctors should work closely with parents and give advice that parents may or may not accept. Wherever possible, a mutually trusting and respectful working relationship should be developed and maintained, both because it will be in the best interests of the child and because it will tend to lead to far better experiences of medical care for all involved.

Also, consider whether your decision would have been the same about performing an extra venepuncture for a special blood test for an ethically approved research project.

Distinction is often made between therapeutic research, where there is an intention to benefit the individual subject, and non-therapeutic research, which carries a wider societal benefit but without intent to benefit individuals. Research that fails to benefit individuals may be ethical provided that it involves an acceptable level of risk.

Where a child suffers from a particular disease, e.g. acute lymphoblastic leukaemia, randomised clinical trials may be used to compare treatment regimens. The ethical justification for such trials is that there is no good reason to believe that one of the treatments would be better than the other – 'therapeutic equipoise' – and that the standard treatment used for comparative purposes is the best currently available.

The situation is different when an investigation, e.g. blood test, X-ray or intervention, is proposed for normal

Case History

5.2 Acute lymphatic leukaemia, truth-telling and stopping treatment

Jane, aged 10 years, has acute lymphoblastic leukaemia which was diagnosed 4 years ago. She has relapsed, with early involvement of the central nervous system. She is well known to the staff of her local children's ward as she has had four relapses of her leukaemia and a previous bone marrow transplant. It is the opinion of her paediatric consultant that no further medical treatment is likely to be curative. Jane asks one of the junior paediatric doctors why her parents had been so upset following a recent discussion with the consultant, at which she had not been present. The parents had made it very clear to all the staff that they did not want their child to be informed of the poor prognosis, nor would they tell her why she was not having further chemotherapy.

The parents have heard of a new drug which is claimed, in some reports on the internet, to help such children. However, it is very expensive, there is evidence that it does not cross the blood–brain barrier and the doctors consider it highly unlikely to be of benefit. The parents insist on a trial of the drug.

Ethical issues to consider are:

- *Autonomy* – the parents claim the right to control the information reaching their child on the grounds that it is in her best interests as judged by them.
- *Truth-telling* – the staff feel that it would be wrong to reassure her falsely.
- *Non-maleficence* – the parents wish to avoid the shock of the news and the loss of hope in their daughter.
- *Beneficence* – the staff wish to support the child effectively, which would be difficult if she were to be isolated by ignorance of what is upsetting her family and carers.
- *Justice* – should scarce resources be used on this new drug? Because her parents are desperate, should Jane be given a drug which, in the specialist's opinion, will not benefit her?
- *Best interests* – what are Jane's best interests and who should decide them? What weight should be given to Jane's own views based on her experience of her illness?

In such situations, further discussion between the parents and staff whom they trust is usually the key to resolving the situation. The parents will need to understand the mutual benefits of adopting as open a pattern of communication as possible. They may be helped by a member of staff being present or helping them talk or listen to the child, who will usually understand more than the parents suspect.

Parents almost always wish to do the best for their child. Detailed explanation is likely to help them see that the child's best interests may not be to seek further cure but to accept a change of focus towards palliative care. A second opinion from an independent specialist may be helpful, as may a specific ethical review. If, despite all efforts to reach agreement, the parents reject the doctor's advice, it is fairest to let a court of law decide whether or not to accept the parents' demands.

children as part of a control group in a trial or for the purpose of establishing a normal range. Both can be ethically justified provided that the procedure in question carries no more risk than generally encountered and accepted in everyday life.

Whatever the nature of the research a number of criteria must be met:

- Appropriate research should be first carried out in adults or older children.
- The project should have a sound scientific basis and be well designed.
- The researchers should be competent to carry it out in the time specified.
- Sufficient information should be given in a form comprehensible to the child and family to enable them to give valid consent to participation, e.g. by provision of information sheets in an appropriate form and language or by the use of independent translators.
- Parents must have the option to withdraw their child from the research at any stage without prejudice.
- The project must be reviewed and approved by an independent scientific and ethical process (Research Ethics Committee).

Summary

Ethics in paediatrics

- Both clinicians and parents aim to do what is in their child's best interests
- Conflicting views can usually be resolved by good communication
- If not resolved – help may be sought from further, wider communication, or a second, truly independent opinion, or sometimes from hospital ethical committees; or may go to court.
- Older children who understand the issues and have strong views as to what should or should not be done to them – there is increasing ethical and legal support for them to exercise as much autonomy as they are capable of.

Evidence-based paediatrics

Clinicians have always sought to make decisions in the best interests of their patients. However, such decisions have often been made intuitively, given as clinical opinion, which is difficult to generalise, scrutinise or challenge. Evidence-based practice provides a systematic approach to enable clinicians to efficiently use the best available evidence, usually from research, to help them solve their clinical problems. The difference between this approach and old-style clinical practice is that clinicians need to know how to turn their clinical problems into questions that can be answered by the research literature, to search the literature efficiently, and to analyse the evidence, using epidemiological and biostatistical rules (Figs 5.8, 5.9). Sometimes, the best available evidence will be a high-quality systematic review of randomised controlled trials, which are directly applicable to a particular patient. For other questions, lack of more valid studies may mean that one has to base one's decision on previous experience with a small number of similar patients. The important factor is that, for any decision, clinicians know the strength of the evidence, and therefore the degree of uncertainty. As this approach requires clinicians to be explicit about the evidence they use, others involved in the decisions (patients, parents, managers and other clinicians) can debate and judge the evidence for themselves.

Why practise evidence-based paediatrics?

There are many examples from the past where, through lack of evidence, clinicians have harmed children, e.g.

- *Blindness from retinopathy of prematurity*. In the 1950s, following anecdotal reports, many neonatal units started nursing all premature infants in additional ambient oxygen, irrespective of need. This reduced mortality, but as no properly conducted trials were performed of this new therapy, it took several years for it to be realised that it was also responsible for many thousands of babies becoming blind from retinopathy of prematurity.
- *Advice that babies should sleep lying on their front (prone), which increases the risk of sudden infant death syndrome (SIDS)*. Medical advice given during the 1970s and 1980s, to put babies to sleep prone, appears to have been based on physiological studies in preterm babies, which showed better oxygenation when nursed prone. Furthermore, autopsies on some infants who died of SIDS showed milk in the trachea, which was assumed to have been aspirated and this was thought to be more likely if they were lying on their back. However, an accumulation of more valid evidence from cohort and case–control studies showed that placing term infants prone was associated with an *increased* risk of SIDS.

Evidence-based medicine allows clinicians to be explicit about the probability (or risk) of important outcomes. For example, in discussing with parents the prognosis of a child who has had a febrile seizure, one can state that 'the risk of developing epilepsy is 1 in 100', instead of using vague terms, such as 'he/she is unlikely to develop epilepsy'.

Explicit analysis of evidence has also become more important with the increasing delivery of healthcare by teams rather than individuals. Each team member needs to understand the rationale for decisions and the probability of different outcomes in order to make their own clinical decisions and to provide consistent information to patients and parents.

To what extent is paediatric practice based on sound evidence?

There are two paediatric specialities in which there is a considerable body of reliable, high-quality evidence underpinning clinical practice, namely paediatric oncology and, to a lesser extent, neonatology. Management protocols of virtually all children with cancer are part of multicentre trials designed to identify which treatment gives the best possible results. The trials are national or, increasingly, international, and include short- and long-term follow-up. Examples of the range of evidence available in paediatrics are given in Box 5.4. In general, the evidence base for paediatrics is poorer than in adult medicine. Reasons for this include:

- The relatively small number of children with significant illness requiring investigation and treatment. To overcome this, multicentre trials are required, which are more difficult to organise and expensive.
- Additional ethical limitations
  - Subjecting children to additional investigations or giving a new treatment is severely limited by the inability of the child to give consent. Some parents are concerned that participating in a trial could mean that their child could receive treatment that turns out to be inferior to the standard treatment and could have unknown side-effects.
  - There is concern over the ability of parents to give truly informed consent immediately after the acute onset of serious illness, e.g. the birth of a preterm infant, meningococcal septicaemia or meningitis.
- There is limited investment by the pharmaceutical industry in drug trials, as drug use in children is insufficient to justify the cost and ethical difficulties of conducting trials. As a result, approximately 50% of drug treatments in children are unlicensed ('off label').

The consequence is that there is less of a culture of randomised controlled trials in paediatrics compared with adult medicine.

For evidence-based practice to become more widespread, clinicians must recognise the need to ask

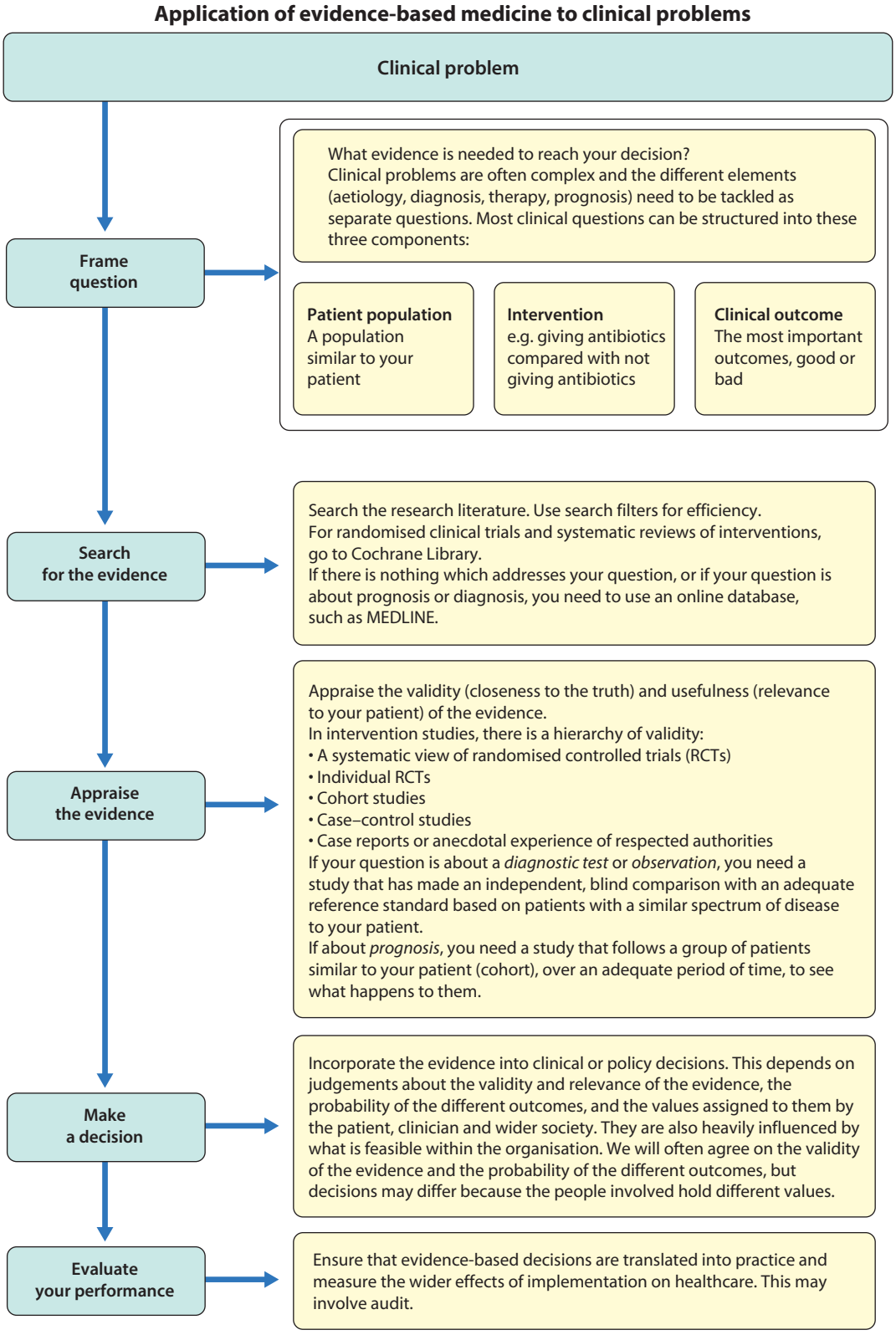


Figure 5.8 Application of evidence-based medicine to clinical problems.

Example of evidence-based practice in solving a clinical problem – the management of acute otitis media

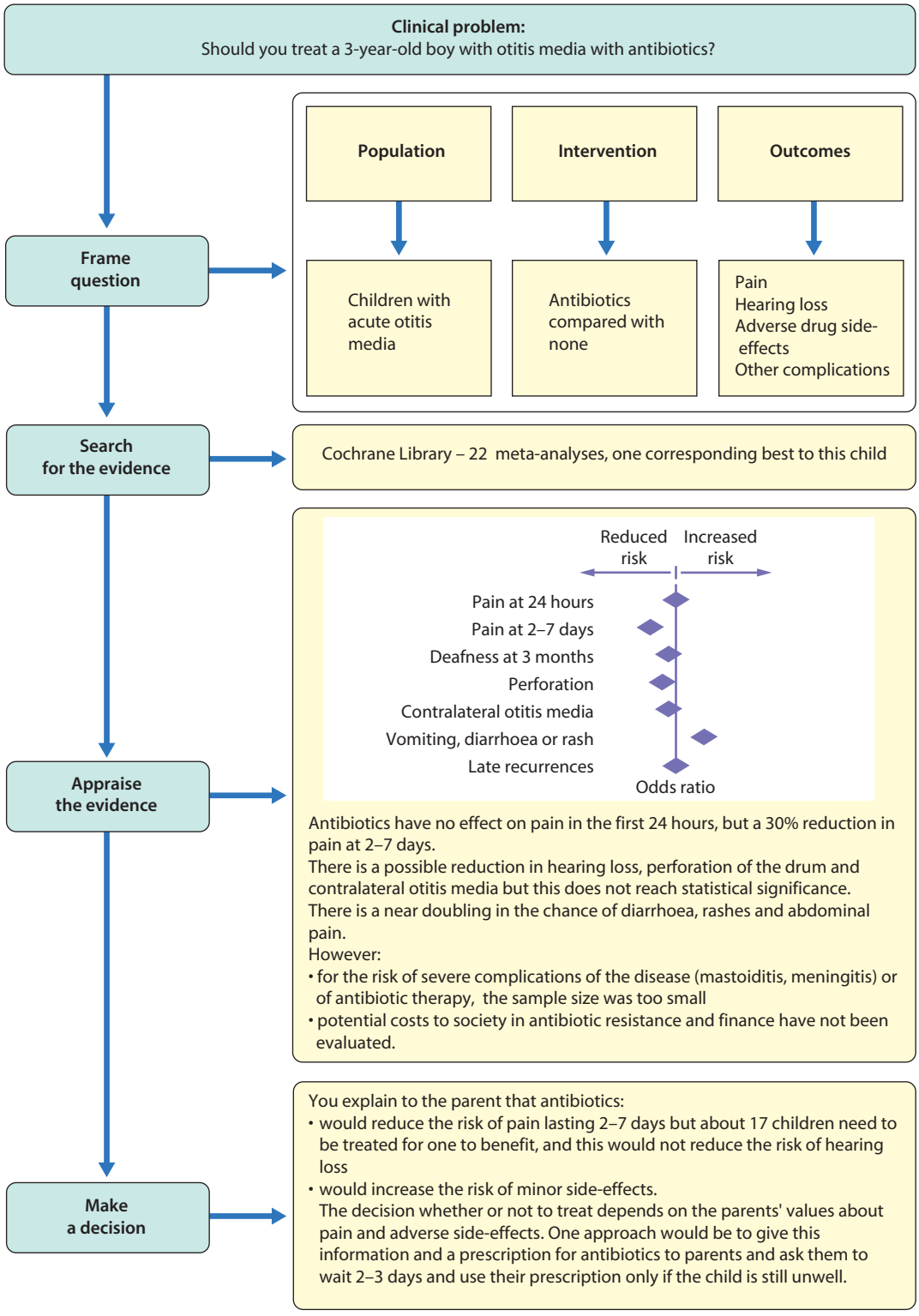


Figure 5.9 An example of an evidence-based medicine approach to a clinical problem – the treatment of acute otitis media with antibiotics. (Sanders S, Glasziou PP, Del Mar C et al. (2004 last updated 2008). Antibiotics for acute otitis media in children. *Cochrane Database of Systematic Reviews* (1) CD000219).

Box 5.4 Examples of the range of evidence available in paediatrics

1. Clear evidence of benefit

Surfactant therapy in pre-term infants

The meta-analysis (see Fig. 10.12) from a Cochrane systematic review shows that mortality is reduced by 40% in preterm infants with respiratory distress syndrome (RDS) treated with surfactant compared with placebo.

This evidence was rapidly produced and introduced into practice as:

- Respiratory distress syndrome is a common cause of death and morbidity in a neonatal intensive care unit
- There is a clearly understood disease mechanism for respiratory distress syndrome, i.e. surfactant deficiency
- The effect of surfactant treatment was immediately obvious at the cot-side – ventilator settings usually have to be reduced shortly after administration
- Potential benefits and side-effects could be clearly defined and identified
- Neonatologists are a relatively small group of doctors who meet regularly – national and international studies could be organised and their results quickly disseminated
- There was financial support and involvement from the pharmaceutical industry.

2. Clear evidence, but need to balance benefits and harms

Antibiotic treatment for children with otitis media

As shown in Figure 5.9, there is a balance of risk and benefits.

3. No clear evidence

Bulk-forming laxatives for constipation

Bulk-forming laxatives, such as methylcellulose or ispaghula husk, are used in children with constipation. However, this is not based on clear evidence. There are no systematic reviews and no randomised controlled studies of these agents in children.

Some possible reasons for the lack of evidence on the use of these laxatives in this common condition are:

- constipation is not a life-threatening disorder
- the causes are multifactorial and the disease mechanism is not clearly defined
- there is a belief that there are likely to be few side-effects to the use of bulk-forming laxatives and clinicians are prepared to prescribe them without clear evidence
- there is limited support for studies from the pharmaceutical industry
- the research agenda is not driven by such clinical problems.

questions, particularly about procedures or interventions which are common practice. However, evidence-based medicine is not cookbook medicine. Incontrovertible evidence is rare, and clinical decisions complex, which is why clinical care is provided by clinicians and not technicians. Evidence-based healthcare cannot change this, but is an essential tool to help clinicians make rational, informed decisions together with their patients. In addition, evidence-based paediatrics provides a way for clinicians to articulate their priorities for research and thereby set a research agenda which is relevant to service needs.

Summary

Evidence-based paediatrics

- requires clinical problems to be framed into questions, to search the literature and then appraise the evidence in order to make a decision
- is less well developed than in adult medicine
- should be adopted whenever possible; however, clinical decisions are complex and the evidence base usually informs rather than determines clinical decision-making.

Further reading

Websites (Accessed May 2011)

CEBM (Centre for Evidence-Based Care, Oxford) Available at: http://www.cebm.net

Every Child Matters – national framework for services for children and young people. Available at: www.education.gov.uk

http://www.tripdatabase.com

http://clinicalevidence.bmj.com/ceweb

http://www.cochrane.org

6

Paediatric emergencies

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There are few situations that provoke greater anxiety than being called to see a child who is seriously ill. This chapter outlines a basic approach to the emergency management of seriously ill children.

The seriously ill child

The rapid clinical assessment of the seriously ill child will identify if there is potential respiratory, circulatory or neurological failure. This should take less than 1 minute. Normal vital signs are shown in Figure 6.1 and how a rapid assessment is performed is shown in Figure 6.2.

Resuscitation is given immediately, if necessary, followed by secondary assessment and other emergency treatment.

The seriously ill child may present with shock, respiratory distress, as a drowsy/unconscious or fitting child or with a surgical emergency. Their causes are listed in Figure 6.4. In children, the key to successful outcome is the early recognition and active management of conditions that are life-threatening and potentially reversible.

Doctors should be able to provide life support for children of all ages, from newborn to adolescents.

Summary

Regarding the seriously ill child

- Prevention of cardiopulmonary arrest is by early recognition and treatment of respiratory distress, respiratory or circulatory failure.

Cardiopulmonary resuscitation

In adults, cardiopulmonary arrest is often cardiac in origin, secondary to ischaemic heart disease. In contrast, children usually have healthy hearts but experience hypoxia from respiratory or neurological failure or shock. If this occurs, irrespective of the cause, basic life support must be started immediately.

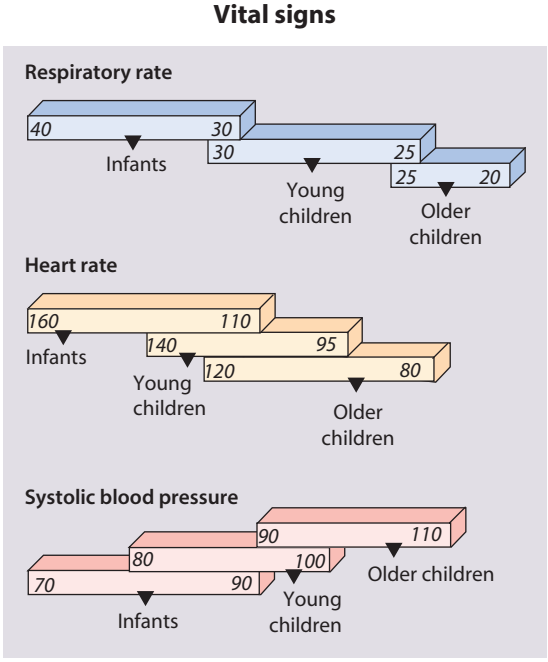


Figure 6.1 Variation in the normal range for respiratory rate, heart rate and systolic blood pressure with age.

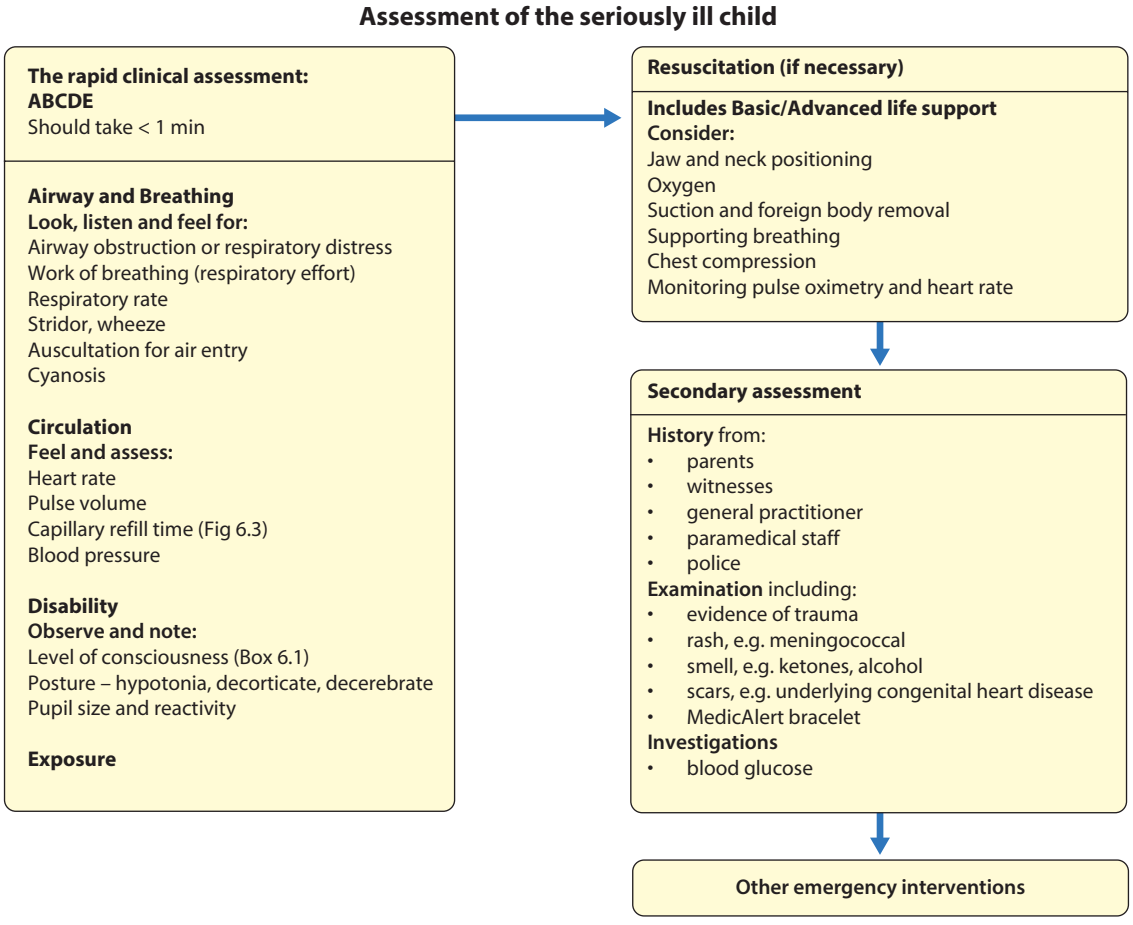


Figure 6.2 Assessment of the seriously ill child.

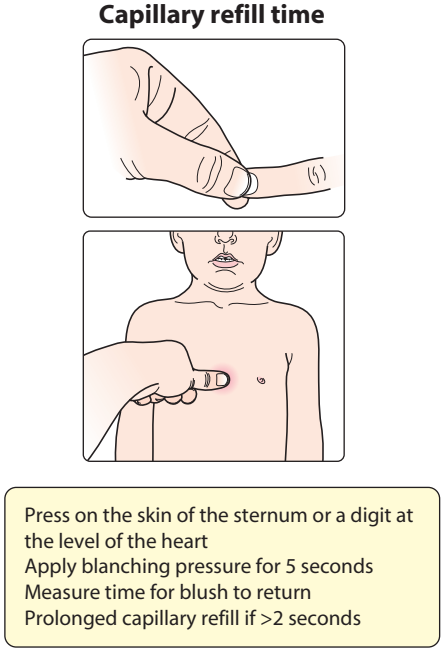


Figure 6.3 Capillary refill time. Digital pressure for 5 seconds. Normal <2 seconds.

Capillary refill time is affected by body exposure to a cold environment.

Box 6.1 AVPU rapid assessment of level of consciousness – more detailed evaluation is with the Glasgow Coma Scale (see Table 6.2)

A	ALERT
V	Responds to VOICE
P	Responds to PAIN
U	UNRESPONSIVE

A score of P means that the child's airway is at risk and will need to be maintained by a manoeuvre or adjunct.

Basic life support

See Figure 6.5.

Advanced life support (Fig. 6.6)

Children who have been resuscitated successfully should be transferred to a paediatric high-dependency or intensive care unit.

The seriously injured child

Management of the seriously injured child must take account of potential injury to the cervical spine and other bones and internal injuries (Fig. 6.7).

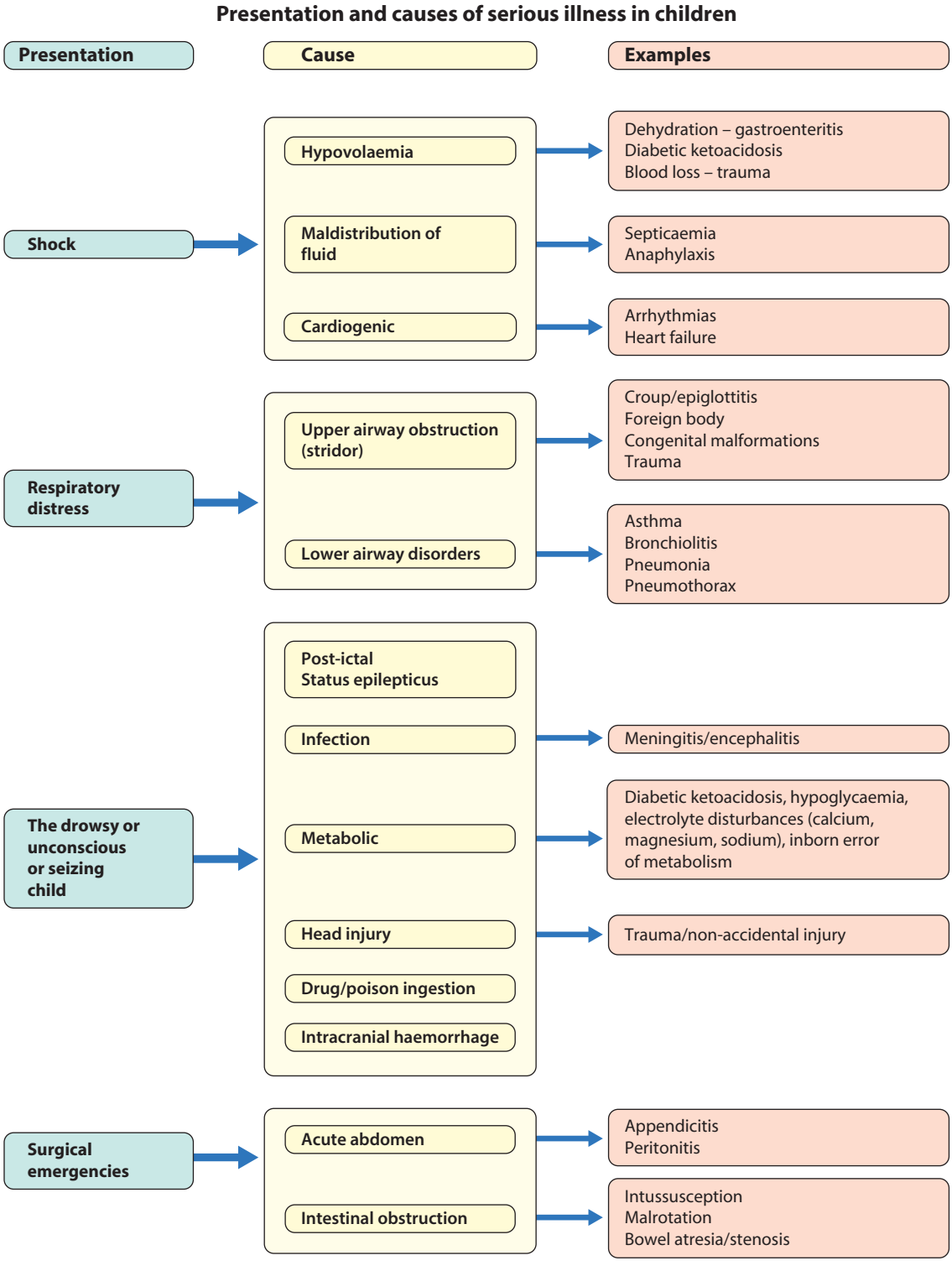


Figure 6.4 The main modes of presentation of serious illness in children and their causes.

Basic life support
(Trained resuscitator, no equipment)

SAFE approach

→ **S**hout for help
Approach with care
Free from danger
Evaluate **ABC**

Check responsiveness:
Ask 'Are you all right?'
Stimulate gently
Do not shake infants or suspected cervical spine injury

No response

Open airway:
- Head tilt, chin lift
- Jaw thrust (if unsuccessful)

No breathing

Check breathing for max 10 s:
- Look – for chest movement
- Listen – for breath sounds
- Feel – for air movement

Breathe
Remove any obvious obstruction
Give 5 initial rescue breaths

Assess 'signs of life' - movements, coughing, normal breathing
Check pulse for max 10 s:
>1 year old – carotid, femoral
<1 year old – brachial, femoral

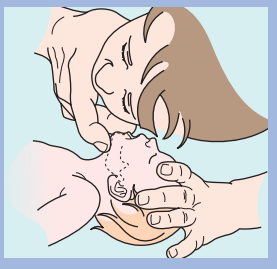
No 'signs of life' (unless definite pulse >60/min)

Compress chest
15 chest compressions: 2 breaths
Rate 100–120 compressions/min
Compression: ventilation ratio for all children:
two rescuers (or lone trained resuscitator) – 15:2
lone rescuer (not trained) – 30:2

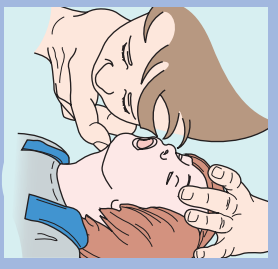
Push 'hard and fast'

What should an untrained bystander do?
Perform chest compressions or expired air ventilation (mouth-to-mouth) – it's better than doing nothing

Airway
Airway opening using head tilt/chin lift manoeuvre



Chin lift in infants
- Head in **neutral** position
- Avoid overextension
- Remove secretions/foreign body under direct vision

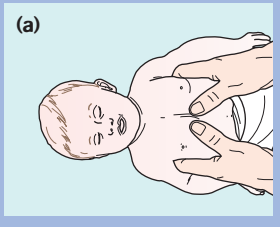


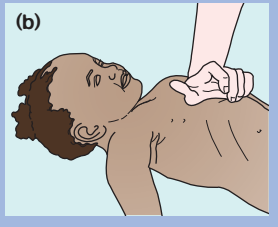
Chin lift in children
Head in 'sniffing' position

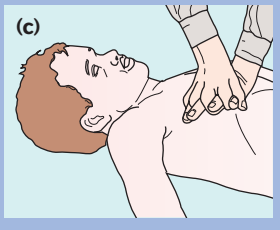
Jaw thrust
Two fingers of each hand behind each side of the mandible and push the jaw forward

Breathing
Infant: Mouth over infant's nose and mouth
Child: Pinch nose, mouth to mouth
The chest should rise with each breath. Blow for 1–1.5 seconds

Circulation
Optimal position for chest compression







(a) Infant. Two thumbs on lower third of sternum with hands round the thorax (needs two rescuers). If alone – compress sternum with tips of two fingers.
(b) Small child. Heel of one hand over lower third of sternum.
(c) Large child. Both hands over lower third of sternum. Depress the sternum by at least one-third of the depth of the chest

Figure 6.5 Basic life support. (Adapted from Resuscitation Council (UK). 2010. Resuscitation Guidelines.)

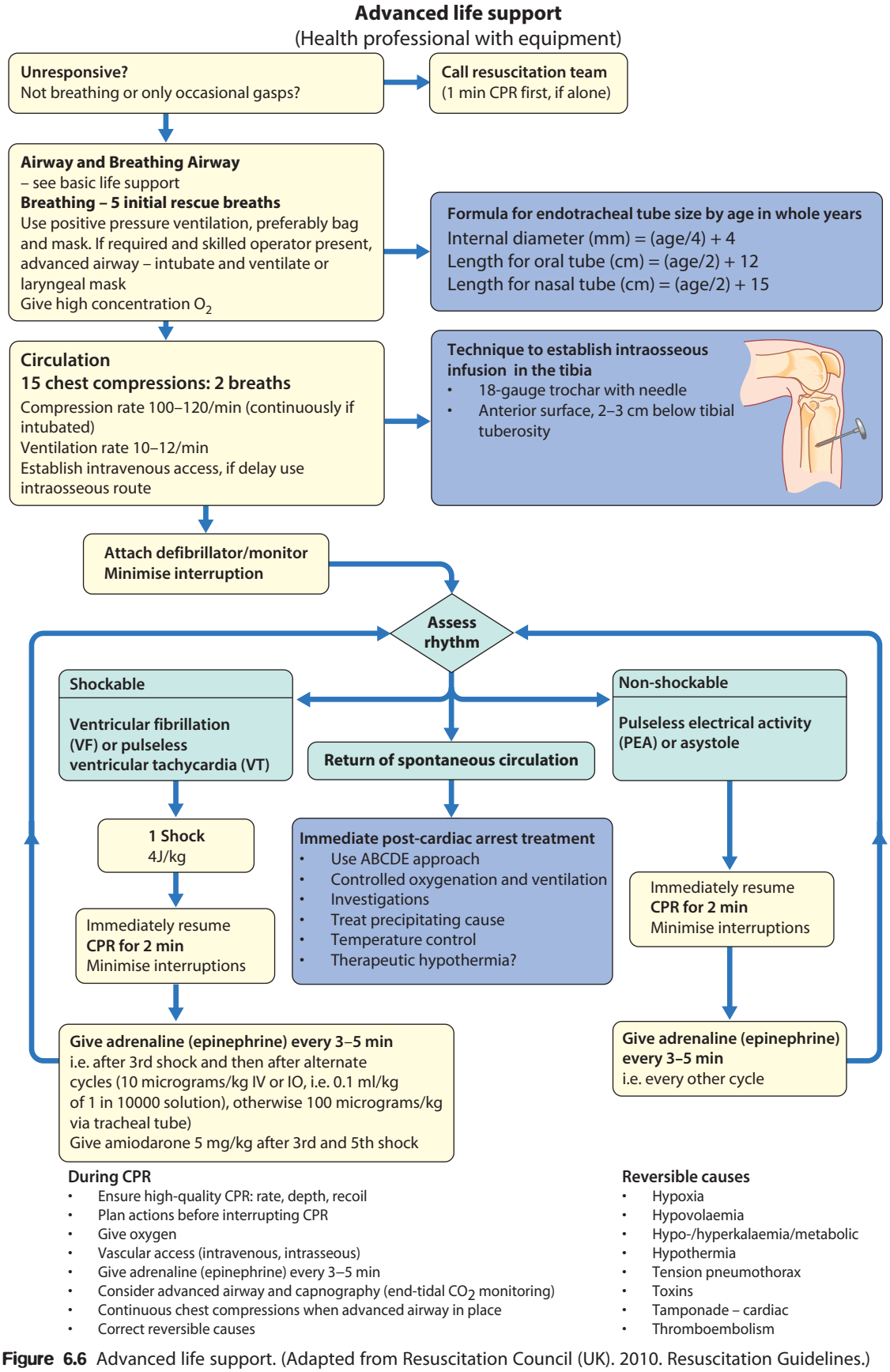


Figure 6.6 Advanced life support. (Adapted from Resuscitation Council (UK). 2010. Resuscitation Guidelines.)